Contents

1 Measures

1.1 Rings

Recall the following definition of a ring.

Definition. (Ring) A collection of subsets A is a ring on E if it contains the empty set and $a, b \in A$ implies $a \cup b \in A$, and $b \setminus a \in A$.

Proposition. The name ring for the above definition actually kind of makes sense

Proof. We need the additional assumption that $E \in A$ for this proof to work so that 1 exists in the ring.

Note that we will consider $b\triangle a = (b \setminus a) \cup (a \setminus b)$ to be $b + a$ and $b \cap a$ to be $b * a$ and we want to prove that these actually satisfy the ring properties.

Note that we consider 1 to be all of E and 0 to be the empty set. It is easy to check that these satisfy the "identity" properties.

Note that $b\triangle a = a\triangle b$ as they are both equal to the set of all things in exactly one of a and b.

Note that

$$b \cap (a\triangle c) = (b \cap a) \triangle (b \cap c)$$

by a truth table of the 8 possibilities of a, b, c and

$$a\triangle (b\triangle c) = (a\triangle b) \triangle c$$

is the set of everything in exactly one of a, b, c and everything in all 3 of a, b, c.

It is easy to check the rest of the ring axioms.

□

1.2 Relation to Topological spaces

Definition. Let E be any topological space and A be the set of open subsets of E, then the Borel σ-algebra is $\mathcal{B}(E) = \sigma(A)$. This is the same definition as for reals but in general topological spaces, and we write $\mathcal{B}$ for $\mathcal{B}(\mathbb{R})$ by convention.

Definition. A measure μ on $(E, \mathcal{B}(E))$ is called a Borel measure.

Definition. A Borel measure μ is said to be outer regular if for any $A \in \mathcal{B}(E)$ we have

$$\mu(A) = \inf_{A \subseteq U,\ \text{U is open}} (\mu(U))$$

Definition. A Borel measure μ is said to be inner regular if for any $A \in \mathcal{B}(E)$ we have

$$\mu(A) = \sup_{K \subseteq A,\ \text{K is compact}} (\mu(K))$$

Definition. If a Borel measure on a Hausdorff topological space has the property that $\mu(K)$ is defined for every compact set K in E and that for each compact set K it is the case that $\mu(K) < \infty$, and it is outer regular and inner regular, it is said to be a Radon measure.

Remark. When it comes to measurable functions, it is just like continuity in topology where the difference is that instead of "pre images of open sets are open" its "pre images of sets in the sigma algebra are in the sigma algebra".

Definition. Let E, F be topological spaces and let $f : E \to F$ be measurable with respect to their Borel σ-algebra. Then f is said to be a Borel function.

Proposition. If E, F are topological spaces and $f : E \to F$ is continuous then f is measurable with respect to borel σ-algebra. We will call the σ-algebras $\mathcal{E}$ and $\mathcal{F}$ respectively.

Proof. Note that $f^{-1}(U)$ is in $\mathcal{E}$ for all U open by definition, and thus by the proposition above it is true for any U that is in the borel σ-algebra. Therefore f is measurable by definition.

□

1.3 The Borel-Cantelli Lemmas

Say we are rolling a dice. Instead of asking what the probability of getting a 6, we might be interested instead in the probability of getting a 6 infinitely often. Intuitively, the answer is "it happens with probability 1", because in each dice roll, we have a probability of $1/6$ of getting a 6, and they are all independent.

We would like to make this precise and actually prove it. We will do this as follows.

Definition. Let A_n be a sequence of events. We define

$$\limsup A_n = \bigcap_{n=1}^{\infty} \bigcup_{m=n}^{\infty} A_m$$

$$\liminf A_n = \bigcup_{n=1}^{\infty} \bigcap_{m=n}^{\infty} A_m$$

Intuitively, $\limsup A_n$ is saying that for every n there is an $m \geq n$ such that A_m occurs, so it is the set of A_m that occurs infinitely often. $\liminf A_n$ is saying there exists an n such that for all $m \geq n$ A_m occurs. i.e. it is the set of outcomes that occur not only infinitely often but in all but a finite number of the events.

Theorem. (Borel-Cantelli lemma) If

$$\sum_{n=1}^{\infty} P\left[A_n\right] < \infty$$

then

$$P\left[A_n \text{ infinitely often}\right] = 0$$

Proof. For each k we have

$$\begin{aligned} P\left[A_n \text{ infinitely often}\right] &= P\left[\bigcap_{n=1}^{\infty} \bigcup_{m=n}^{\infty} A_m\right] \\ &\leq P\left[\bigcup_{m=k}^{\infty} A_m\right] \\ &\leq \sum_{m=k}^{\infty} P\left[A_m\right] \\ &\to 0 \end{aligned}$$

where the limit is taken as $k \to \infty$. Therefore the probability is less than every element of a sequence that approaches 0 so it must be 0.

$\square$

Remark. In the above proof we can replace P with an arbitrary measure μ and it will still work. It is therefore true that the above theorem holds in arbitrary measure spaces - If we have a countably infinite family of sets with finite total measure then their intersection has measure 0.

Theorem. (Borel-Cantelli lemma II) If

$$\sum_{n=1}^{\infty} P\left[A_n\right] = \infty$$

and A_n are independent events then

$$P\left[A_n \text{ infinitely often}\right] = 1$$

Note that we need independence because otherwise we could flip a coin and set all A_n to "get a heads" in which case $\sum_{n=1}^{\infty} P\left[A_n\right] = \sum_{n=1}^{\infty} \frac{1}{2} = \infty$ but $P\left[A_n \text{ infinitely often}\right] = \frac{1}{2}$.

By setting A_n to "roll a 6" this will make the dice intuition rigorous. This is similar to some stuff we did in the Markov chains course.

Proof. Recall that if A_n is independent as a sequence so is A_n^C, as this is easy to check from the definition. Then we have, using $1 - x \leq e^{-x}$ and independence, for each n and taking $N \to \infty$,

$$P\left[\bigcap_{m=n}^{N} A_m^C\right] = \prod_{m=n}^{N} P\left[A_m^C\right]$$

$$= \prod_{m=n}^{N} \left(1 - P\left[A_m\right]\right)$$

$$\leq \prod_{m=n}^{N} e^{-P[A_m]}$$

$$= e^{-\sum_{m=n}^{N} P[A_m]}$$

$$\to 0$$

Therefore for all n we have

$$P\left[\bigcap_{m=n}^{\infty} A_m^C\right] = 0$$

So by countable subadditivity we have

$$P\left[\bigcup_{n=1}^{\infty}\bigcap_{m=n}^{\infty} A_m^C\right] = 0$$

Therefore

$$P\left[\bigcap_{n=1}^{\infty}\bigcup_{m=n}^{\infty} A_m\right] = 1 - P\left[\bigcup_{n=1}^{\infty}\bigcap_{m=n}^{\infty} A_m^C\right] = 1$$

Since the LHS is the probability of A_m infinitely often we are done.

□

2 Measurable functions

2.1 Constructing new measures

As a reminder a Radon measure is finite on compact sets, inner regular and outer regular. We will now prove a stronger version of a theorem we proved in Level 6.2.

Theorem. Let $g : \mathbb{R} \to \mathbb{R}$ be non-constant, non-decreasing and right continuous. Then there exists a unique Radon measure dg on $\mathcal{B}$ such that $dg\left((a, b]\right) = g\left(b\right) - g\left(a\right)$ and we obtain all non-zero Radon measures on $\mathbb{R}$ this way.

Proof. Take I and f as in the previous lemma and let μ be the restriction of the Lebesgue measure to Borel subsets of I. Now f is measurable since it is left continuous (so pre-images of intervals, the generating set, are intervals). Now we define dg such that

$$dg\left((a, b]\right) = \mu\left(\{x \in I : a < f\left(x\right) \leq b\}\right) = \mu\left(\{x \in I : g\left(a\right) < x \leq g\left(b\right)\}\right) = g\left(b\right) - g\left(a\right)$$

The Caratheodory extension theorem extends this to a measure on all Borel sets.

Therefore dg is a radon measure with the required property - note that it is inner and outer regular by the proof of the Caratheodory extension theorem. For outer regularity, we need to extend to open sets. For the normal measure it was fine - the justification was "add a bit on the end". Here this still works because everything is right continuous. Inner regularity follows from outer regularity of the complement.

The same proof that showed that the Lebesgue measure is unique works to show that dg is unique.

We can get all non-zero radon measures this way: Given a non-zero radon measure ν define a function g by

$$g(y)=\begin{cases}-\nu((y,0]) & y<0\\ \nu((0,y]) & y>0\\ 0 & y=0\end{cases}$$

Then it is easy to check that for any $a<b$ in the various cases we have $\nu((a,b])=g(b)-g(a)$. Since ν is non-zero g is non-constant. It is easy to see that g is non-decreasing.

Note that a measure has the property that the measure of an intersection of a decreasing sequence of sets or a union of an increasing sets is the limit of the measure. It is therefore the case that g is right-continuous as if $y\searrow 0$ then the intersection is empty so the measure approaches 0, and if $c\neq 0$ and $c_n\searrow c$ then $g(c_n)\searrow g(c)$ by this fact and by definition of g. Therefore g is right continuous.

□

2.2 Random variables

Let (Ω,F,P) be a probability space and $(E,\mathcal{E})$ be a measurable space. Then we can define an E-valued random variable by any measurable function $X:\Omega\to E$.

This seems like a rather weird definition.

Example. We could make the random variable "1 if a coin is heads 0 otherwise" by letting X be the function $X:\Omega(H,T)\to\mathbb{R}$ by $X(H)=1$ and $X(T)=0$.

Example. We could make a standard normal distribution by setting $\Omega=\mathbb{R}$, $F=\mathcal{B}$, and P defined by the measure above where our function g is given by the cdf of a standard normal. Then we would set $E=\mathbb{R}$ and X to be the identity function.

Definition. (Distribution/Law) Given a random variable $X:\Omega\to E$ we say that the distribution or law of X is the image measure μ_X defined by

$$P(X\in A)=\mu_X(A)=P\left(X^{-1}(A)\right)$$

If $E=\mathbb{R}$ then we define

$$F_X(x)=\mu_X((-\infty,x])=P(X\leq x)$$

This is the usual cumulative distribution function, and from properties of measures (ie, continuity from below: That the measure of unions is the limit of the measure) we see that as $x\to\infty$ $F\to 1$ and that as $x\to-\infty$ $F\to 0$. F is also non-decreasing and right continuous and any such F with the right values at $\pm\infty$ gives a real random variable. With the same proof that we get a radon measure from such a function, such a function gives a random variable.

Note that the construction is useful in practice. As we briefly discussed at the end of the IA probability course, if we can make a computer generate a $U[0,1]$ variable then given another variable with cdf F we take its generalized inverse g, generate a uniform random number x and then calculate $g(x)$.

Note that we want to say that random variables X,Y are said to be independent if for any A,B in $\mathcal{E}$ (in fact any A,B in any π-system that generates $\mathcal{E}$) we have

$$P[X\in A,Y\in B]=P[X\in A]\,P[Y\in B]$$

But note that for each fixed X the events $X\in A$ for different subsets A form a π-system (easy to check) and similarly for Y and B. By a theorem in section 1.4 this statement just says that the σ-algebras generated by these π-systems are independent.

We therefore define independence as follows:

Definition. A sequence of random variables X_n is said to be independent if the σ-algebras $\sigma(X_n)$ are independent.

We have now managed to generalize independence beyond the case where there is a density or the variables are discrete. We will prove several theorems that show that independence by this definition coincides with the properties we intuitively expect independence to satisfy.

Proposition. Two real-valued random variables X, Y are independent if and only if for all x, y we have

$$P[X \leq x, Y \leq y] = P[X \leq x] P[Y \leq y]$$

In fact if we have a countable sequence of random variables $X_1, X_2, ...$ then they are independent if for every n and $x_1, x_2, x_3, ..., x_n$ and increasing sequence of natural numbers we have

$$P[X_{a_1} \leq x_{a_1}, X_{a_2} \leq x_{a_2}, ..., X_{a_n} \leq x_{a_n}] = P[X_{a_1} \leq x_{a_1}] P[X_{a_2} \leq x_{a_2}] ... P[X_{a_n} \leq x_{a_n}]$$

Proof. We actually just need to show that if we have this factorization then the variables are independent. That direction is obvious because $(-\infty, x]$ is a generating π-system for Borel sets of $\mathbb{R}$.

As for the second part of the claim, this is by our original definition of independence of σ-algebras and the first part of the claim.

□

Lemma. This is a fact which we have probably used in the past as it seems obvious but it is surprisingly difficult to actually prove. There exists a sequence R_n of independent Bernoulli(1/2) random variables.

Proof. Let $(\Omega, F, P) = ((0,1], \mathcal{B}, \text{Lebesgue})$. Given $\omega \in \Omega$ write $\omega = \sum_{n=1}^{\infty} \omega_n 2^{-n}$, ie its binary expansion. We will disallow infinite sequences of zeroes to make it unique. You can probably see now what the idea is here.

Define $R_n(\omega) = \omega_n$. Then these are each measurable as we can write them as a finite sum of indicator functions as follows:

$$R_1(\omega) = 1_{\left(\frac{1}{2}, 1\right]}(w)$$

$$R_2(\omega) = 1_{\left(\frac{1}{4}, \frac{1}{2}\right]}(w) + 1_{\left(\frac{3}{4}, 1\right]}(w)$$

$$R_3(\omega) = 1_{\left(\frac{1}{8}, \frac{1}{4}\right]}(w) + 1_{\left(\frac{3}{8}, \frac{1}{2}\right]}(w) + 1_{\left(\frac{5}{8}, \frac{3}{4}\right]}(w) + 1_{\left(\frac{7}{8}, 1\right]}(w)$$

and extend the obvious way to higher n. Then each R_n is a random variable and in fact it is a Bernoulli(1/2) variable as it is the indicator of a measure 1/2 set so it remains to show that they are independent. This is easy because for any finite set of natural numbers $\{a_1, a_2, a_3, ..., a_n\}$ we have that the measure of the intersection of all R_{a_n}'s is 2^{-n}, since it consists of one of the 2^n equally likely possibilities for the binary digits in the a_i positions. Therefore independence follows

□

Lemma. (Grouping lemma) Let X_i be a collection of mutually independent random variables of a probability space and let I_j for $j \in J$ be a collection of pairwise disjoint subsets of I. Then the σ-algebras given by $\sigma(H_j)$, where $H_j = \bigcup_{i \in I_j} \{X_i^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\}$, are mutually independent.

Proof. For each $j \in J$, let P_j be the collection of all events that depend only on finitely many variables in I_j, ie events that can be written as

$$A = \bigcap_{k=1}^{m} \{X_{i_k} \in B_k\}$$

for borel sets B_k. Then this is a π-system (closed under finite intersections and has the empty set) and it generates the σ-algebra $\sigma(H_j)$.

Therefore if we can prove that these π-systems are independent then we will be done by a previous theorem. Specifically, for any finite subset $K \subseteq J$ and any choice of events $A_k \in P_k$ for each $k \in K$ we will prove that we have

$$P\left(\bigcap_{k\in K} A_k\right) = \prod_{k\in K} P\left(A_k\right)$$

To do this note that by definition of our π-systems we can write

$$A_k = \bigcap_{m=1}^{n_k} \left\{X_{i_{k,m}} \in B_{k,m}\right\}$$

where $i_{k,m} \in I_k$

Substituting this into our probability gives

$$P\left(\bigcap_{k\in K} A_k\right) = P\left(\bigcap_{k\in K}\bigcap_{m=1}^{n_k} \left\{X_{i_{k,m}} \in B_{k,m}\right\}\right)$$

Because the blocks I_k are pairwise disjoint we have a finite intersection of events corresponding to distinct variables from the original family. By independence of the original family we have

$$P\left(\bigcap_{k\in K} A_k\right) = \prod_{k\in K}\prod_{m=1}^{n_k} \left\{X_{i_{k,m}} \in B_{k,m}\right\} = \prod_{k\in K} P\left(A_k\right)$$

This is exactly what we wanted to prove.

□

Theorem. There exists an independent sequence for any distribution, and they don't even have to be identical.

Specifically, given any sequence F_n of distribution functions there is a sequence X_n of independent random variables with cdf F_n.

Proof. Fix $m : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ to be any bijection and define $Y_{k,n}$ to be $R_{m(k,n)}$ where R is defined as in the lemma about Bernoulli variables. Now define $Y_n = \sum_{k=1}^{\infty} 2^{-k} Y_{k,n}$. Then Y_n is a sequence of independent random variables that is uniform on $(0, 1)$.

Note that the Y_n are independent by the grouping lemma. Specifically, each Y_n is G_n-measurable with respect to a σ-algebra G_n where the sequence G_1,G_2,G_3,... is mutually independent by the grouping lemma, and thus so are the random variables Y_n.

Now let G_n be the generalized inverse of F_n and define $X_n = G_n\left(Y_n\right)$. Then X_n is a sequence of independent random variables with the prescribed cdf F_n.

□

Now here is a funny fact: The strong law of large numbers, which we will prove later, implies that the probability is 1 that the proportion of 1's in the binary expansion of a uniform $(0, 1)$ variable is a sequence converging to 1/2. The funny thing is though, that for some random number like π, while it passes every statistical test that this is the case and we have no reason to believe it isn't, nobody has managed to prove it. How would one go about proving something like that anyway?

2.3 Convergence of measurable functions

We will discuss a few different notions of convergence.

Definition. Suppose that $(E, \mathcal{E}, \mu)$ is a measure space, and that (f_n) is a sequence of measurable functions and f is a measurable function, each from $E \to \mathbb{R}$. We say that $f_n \to f$ almost everywhere if the set of x such that f_n does not tend to f as a sequence is a measure 0 set. We call this almost sure convergence.

Since $\limsup |f_n - f|$ is measurable as we have seen earlier, it follows that the set of points where this is > 0 is actually a measurable set.

Definition. Suppose that $(E, \mathcal{E}, \mu)$ is a measure space, and that (f_n) is a sequence of measurable functions and f is a measurable function. We say that $f_n \to f$ in measure if for each $\varepsilon > 0$ we have

$$\mu\left(\{x \in E : |f_n(x) - f(x)| \geq \varepsilon\}\right) \to 0 \text{ as } n \to \infty$$

In the context of a probability space this is called convergence in probability, as we saw in the IA probability course, if we consider f_n, f as functions corresponding to random variables.

After we define Lebesgue integration we will be able to define a norm of a function f as follows: Given $p > 0$ we define

$$\|f\|_p = \left(\int |f(x)|^p \, dx\right)^{\frac{1}{p}}$$

Then if $\|f - f_n\|_p \to 0$ then it becomes smaller than ε for each ε which will imply that $|f - f_n(x)| < \varepsilon$ except on a set of measure $< \varepsilon$. Thus this convergence implies convergence in measure.

However, convergence in measure and almost sure convergence do not always imply eachother, however we can still say some useful things.

Theorem. We have the following

i) If $\mu(E) < \infty$ then almost sure convergence implies convergence in measure

ii) For any E, if $f_n \to f$ in measure then f_n has a subsequence that converges to f almost surely.

Proof. (i) can be proven with the same proof as we gave in IA probability (see level 8.4). We use finiteness when we implicitly think that the probability of something approaching 1 means the probability of its complement approaches 0 - if the total measure of the space was something infinite this would be nonsense. We also proved convergence to 0, but thats fine as we can just do everything on $f - f_n$.

Note also that the finiteness is necessary because a counterexample would be $f_n = 1_{[n,n+1]}$ which converges to 0 almost everywhere but *not* in measure, since if $\varepsilon = 0.5$ then $f_n > \varepsilon$ on a set of measure 1 always.

So it remains to prove (ii)

By convergence in measure we can pick a subsequence n_k such that

$$\mu\left(\left\{x \in E : |f_{n_k}(x) - f(x)| > \frac{1}{k}\right\}\right) \leq 2^{-k}$$

Then we have, by countable subadditivity

$$\sum \mu\left(\left\{x \in E : |f_{n_k}(x) - f(x)| > \frac{1}{k}\right\}\right) \leq 1 < \infty$$

Now we can apply the first Borel-Cantelli Lemma. We know that the probability is 0 that $|f_{n_k}(x) - f(x)| > \frac{1}{k}$ infinitely often. This is true for every k so we have almost sure convergence.

$\square$

Definition. If X_n,X are random variables then we say $X_n \to X$ in distribution if $F_{X_n} \to F_X$ at each point where F_X is continuous. We've stated this in levels 6 and 8 so this is just a reminder.

The reason we only care about continuity points is because as we actually saw in level 6, there are only countably many discontinuity points since the cdf is increasing, so its value at all such points determines the distribution.

Example. Let X_n be 1 with probability $1/n$ and 0 otherwise and let them all be independent. Then by the second Borel-Cantelli Lemma X_n is 1 infinitely often with probability 1 (as the sum of $1/n$ is ∞), so X_n does not tend to 0 almost surely, but it does tend to 0 in distribution.

However, there is something we can say on this note.

Before I state the theorem, I will remark on the example above that if we generate a random X in $(0, 1)$ then define a non-independent sequence X_n that is 1 if $X < \frac{1}{n}$ and 0 otherwise, then these are the same distributions but they are not independent, and they actually converge almost surely to 0. We will show that if we have convergence in distribution we can find a probability space and make almost sure convergence happen.

Theorem. (Skorokhod representation theorem of weak convergence)

i) If $(X_n), X$ are defined on the same probability space and $X_n \to X$ in probability then $X_n \to X$ in distribution.

ii) If $X_n \to X$ in distribution then there exists random variables $\left(\tilde{X}_n\right)$ and $\tilde{X}$ defined on a common probability space with $F_{\tilde{X}_n} = F_{X_n}$ and $F_{\tilde{X}} = F_X$ such that $\tilde{X}_n \to \tilde{X}$ almost surely.

Proof. For (i)

Let S be the set of continuity points of F_X. Then given x in S we need to show that $F_{X_n}(x) \to F_X(x)$. For each $\varepsilon > 0$ we can find a δ (by continuity) such that

$$F_X(x - \delta) \geq F_X(x) - \frac{\varepsilon}{2}$$

$$F_X(x + \delta) \leq F_X(x) + \frac{\varepsilon}{2}$$

We fix N large enough such that $n \geq N$ implies that $P[|X_n - X| \geq \delta] \leq \frac{\varepsilon}{2}$, possible by convergence in probability. Then

$$F_{X_n}(x) = P[X_n \leq x] = P[(X_n - X) + X \leq x]$$

We notice that $(X_n - X) + X \leq x$ means that either $X \leq x + \delta$ or $|X_n - X| > \delta$ therefore we can write

$$\leq P[X \leq x + \delta] + P[|X_n - X| \geq \delta]$$

$$\leq F_X(x + \delta) + \frac{\varepsilon}{2}$$

$$\leq F_X(x) + \varepsilon$$

By the same argument we have $F_{X_n}(x) \geq F_X(x) - \varepsilon$ so $F_{X_n}(x) \to F_X(x)$

for (ii)

Set $(\Omega, F, P) = ((0, 1), \mathcal{B}, \text{Lebesgue})$ as usual. Define $\tilde{X}_n(\omega)$ for $w \in (0, 1)$ as our generalized inverse for F_{X_n} and do the same to define $\tilde{X}(\omega)$. This is basically what I did with the above example where I motivated the theorem. We will show that if we do this we indeed get almost sure convergence.

Note that $\tilde{X}$ is an increasing function from $(0, 1) \to \mathbb{R}$ so it only has countably many discontinuities (by general ideas we have seen many times in earlier levels). Therefore we are at a discontinuity point with probability 0, since countable sets have measure 0.

We will now show that for all continuity points $\tilde{X}_n \to \tilde{X}$ (a measure 1 set which we will call Ω_0). This will mean that on this probability space, we satisfy the conditions of the theorem.

We will now fix $\omega \in \Omega_0$ and $\varepsilon > 0$. We want to show that $\left|\tilde{X}_n(\omega) - \tilde{X}(\omega)\right| < \varepsilon$ for n large enough. Note that S has a countable complement so is dense in $\mathbb{R}$ and therefore we can find x^- and x^+ within ε of eachother such that $x^- < \tilde{X}(\omega) < x^+$.

We will pick $\omega^+ \in (\omega, 1)$ such that $\tilde{X}(\omega^+) < x^+$ (this is possible by going sufficiently close to ω as $\tilde{X}$ is left continuous) so now we have

$$x^- < \tilde{X}(\omega) \leq \tilde{X}\left(\omega^+\right) < x^+$$

Then by properties of the generalized inverse we have

$$F_X\left(x^-\right) < \omega < \omega^+ \leq F_X\left(x^+\right)$$

Note that ω^+ was introduced because generalized inverse properties alone would only give us $F_X(x^-) < \omega \leq F_X(x^+)$ which would make the next step invalid.

By convergence in distribution, for sufficiently large n we have

$$F_{X_n}\left(x^-\right) < \omega < F_{X_n}\left(x^+\right)$$

Therefore using properties of the generalized inverse we have

$$x^- < \tilde{X}_n(\omega) \leq x^+$$

Since x^- and x^+ were defined to be within ε of eachother we are done.

□

2.4 Tail events

Recall that if we have a random variable X then we define $\sigma(X)$ to mean that X is the set of pre-images of sets in the σ-algebra of the codomain of X (often the Borel on the reals), which we usually consider to be the Borel σ-algebra on $\mathbb{R}$. We now define a σ-algebra as follows.

$$T_n = \sigma\left(X_{n+1}, X_{n+2}, ...\right)$$

Definition. The tail σ-algebra is defined to be $T = \bigcap_n T_n$.

It is easy to check that intersections of σ-algebras are σ-algebras, it is just that they are closed under the right stuff.

The idea is that T-measurable events depend only on the asymptotic behavior of the X_n's.

Example. Let X_n be a sequence of real random variables, then consider the following

$$\limsup_{n\to\infty} \frac{1}{n}\sum_{j=1}^{n} X_j \text{ and } \liminf_{n\to\infty} \tfrac{1}{n}\textstyle\sum_{j=1}^{n} X_j$$

Then these are both T-measurable.

The reason is that for each k, x the event $\liminf_{n\to\infty} \frac{1}{n}\sum_{j=1}^{n} X_j \leq x$ can be determined entirely by $\{X_{k+1}, X_{k+2}, ...\}$ and so the event is in the pre-image of the σ-algebra from T_k. Since this is true for all k it is in the pre-image of that from T so it is T-measurable. The above are T-measurable random variables.

Finally the set of sequences such that $\lim_{n\to\infty} \frac{1}{n}\sum_{j=1}^{n} X_j$ exists is T-measurable as it is just when the above 2 values are equal.

Theorem. (Kolmogorov 0-1 law) Let X_n be a sequence of independent real valued random variables and let $A \in T$. Then $P(A)$ is either 0 or 1. Furthermore, if X is a T-measurable random variable then there exists a constant c such that $P[X = c] = 1$.

Proof. We will do something funny. We will prove that if $A \in T$ then A is independent of itself, so that $P[A] = P[A \cap A] = P[A]^2$

We will do this by proving that T is independent of T, since events are independent if and only if their σ-algebras are independent.

Let $F_n = \sigma(X_1, X_2, ..., X_n)$. This σ-algebra is generated by the π-system of events of the form

$$A = \{X_1 \leq x_1, X_2 \leq x_2, ..., X_n \leq x_n\}$$

Similarly, T_n can be generated by the π-system of events of the form

$$B = \{X_{n+1} \leq x_{n+1}, X_{n+2} \leq x_{n+2}, ..., X_{n+k} \leq x_{n+k}\}$$

where k can be natural number/positive integer, and we can get this to 'infinite' tails by taking intersections over values of k.

Since X_n is a sequence of independent random variables we know that for any such A, B that A, B are independent. Therefore, F_n is independent of T_n.

Since T is a subset of each T_n it follows (from the definition of independence of σ-algebras) that F_n is independent of T.

Now $\cup_k F_k$ is a π-system which generates the σ-algebra $F_\infty = \sigma(X_1, X_2, ...)$.

Note that the union of σ-algebras is just a collection that contains more subsets - ie any subset that is in any of the σ-algebras, so a finer partition of the starting set.

Now any A in $\cup_k F_k$ has to be in some F_n for n finite, and it is therefore independent of T. By a theorem from sections 1.4 about independence on generating π-systems it follows that F_∞ is independent of T. But now the funny part is that T is *in* F_∞ so T is independent of T.

Now suppose that X is a T-measurable real random variable. Then for each x, $P[X \leq x]$ is either 0 or 1 for all real x.

Now there exists an infimum over x such that this is 1, then this is the c we want.

□

3 Integration

3.1 Definition of the Lebesgue integral

Now that we understand measurable functions we will try to understand how to actually measure them. The notation we will often use will be $\mu(f)$ or $\int_E f d\mu$. We by convention write dx instead of $d\mu$ if we are working with the Borel/Lebesgue measure on $\mathbb{R}$. Also, if μ is a probability measure then this integral will be written as $E[f(x)]$, ie the expectation of $f(x)$.

Note that Lebesgue measurable real functions are those that are measurable when we are going from $\mathbb{R}$ with the Lebesgue measure to $\mathbb{R}$ with the Borel measure. This allows indicator functions of non-Borel measurable sets to be measurable and makes it so we only have to check what happens on intervals to get our function to be measurable. Of all 4 combinations of Lebesgue/Borel in the domain and codomain this actually allows the *most* functions since it makes the pre-image of the "least" things allowed to be the "most" things, and it is easy to check as we only need to check intervals.

Definition. A simple function is a measurable function that can be written as a finite non-negative linear combination of measurable sets. Specifically, any simple function f is of the form

$$f = \sum_{k=1}^{n} a_k 1_{A_k}$$

for $A_k \in \mathcal{E}$ and $a_k > 0$.

Note that simple functions are exactly those that are measurable, non-negative and only take on finitely many values.

We will define the integral of a simple function defined as above to be $\sum_{k=1}^{n} a_k \mu(A_k)$

Note that this is well defined by basic properties of the measure.

For arbitrary non-negative measurable functions we define their integral as follows:

$$\mu(f) = \sup\{\mu(g) : g \leq f, g \text{ is simple}\}$$

Note the following remark: When we talk about assigning a measure to a function this is a function to the reals unless stated otherwise. It may also be to the *extended* reals where we allow $\pm\infty$ to be part of the codomain (note that the single infinity point has measure 0).

For arbitrary measurable functions f we find their integral by defining it to be $\mu(f \wedge 0) - \mu((-f) \wedge 0)$, ie considering the positive and negative parts.

Definition. We say f is integrable if $\mu(|f|)$ is finite.

If we are integrating over a subset of the reals then we call this the Lebesgue integral.

Theorem. Let $f : [a, b] \to \mathbb{R}$ be Riemann integrable. Then it is Lebesgue integrable and the integrals agree.

We will revisit this when we define a measure on $\mathbb{R}^n$. The proof adapts to this case pretty much immediately.

Proof. **Step 1.** We will first prove that f is measurable. To do this we will use a theorem we proved in level 8.2 that says that if f is bounded on a bounded domain (standard conditions for Riemann integrability - and this theorem is not generally true for improper integrals, however it is true if the integral of the absolute value is finite, this is immediate by the dominated convergence theorem where soon we will see the version for Lebesgue integrals), then it is Riemann integrable if and only if it is discontinuous on a measure 0 set.

We will show that for any $U \subseteq \mathbb{R}$ its preimage is measurable. Let C be the set of continuity points and D the set of discontinuity points. Then

$$f^{-1}(U) = \left\{\left\{f^{-1}(U) \cap C\right\} \cup \left\{f^{-1}(U) \cap D\right\}\right\}$$

Note that $\left\{f^{-1}(U) \cap D\right\}$ has measure 0 so we just need to show that $\left\{f^{-1}(U) \cap C\right\}$ is measurable.

Let x be in $f^{-1}(U) \cap C$. Because U is open, there exists some $\varepsilon > 0$ such that $(f(x) - \varepsilon, f(x) + \varepsilon) \subseteq U$.

Now by continuity of f at x this implies that for some δ the δ-neighbourhood of x is in $f^{-1}(U)$. Let V be the union of all these neighbourhoods from the x's in $f^{-1}(U) \cap C$. Then V is open in $[a, b]$ and is therefore measurable. Clearly, $V \cap C$ contains all points in $f^{-1}(U) \cap C$ by definition. To see that the other side of the inclusion holds, note that if $y \in f^{-1}(U) \cap C$ then it is in one of the neighbourhoods from earlier of some x, and these neighbourhoods are all in $f^{-1}(U)$ so $V \subseteq f^{-1}(U)$ and so $V \cap C \subseteq f^{-1}(U) \cap C$.

Since V is open, and C and $f^{-1}(U) \cap D$ are lebesgue measurable we have

$$f^{-1}(U) = \left\{\left\{f^{-1}(U) \cap C\right\} \cup \left\{f^{-1}(U) \cap D\right\}\right\} = \left\{V \cup \left\{f^{-1}(U) \cap D\right\}\right\}$$

so $f^{-1}(U)$ is measurable.

Step 2. We assume for now that f is non-negative. We note that f is Riemann integrable hence bounded by some M, and so no simple function will have more measure then $M(b-a)$. Therefore the integral will be finite and still exist.

To show it equals the riemann integral we use the fact that if $a \leq b$ are simple functions then $\mu(a) \leq \mu(b)$. This is pretty easy to show, by considering the simple function $b-a$ and noting that the integral of simple functions is additive.

Let the riemann integral of f be R.

Now for every natural number n, we use Riemann integrability to make a partition and a rectangle-function overestimating f with integral between R and $R+\frac{1}{n}$. Notice, this is a simple function, and any simple function underestimating f will therefore be $\leq R+\frac{1}{n}$ by the fact above. On the other hand, the simple function that we get from riemann-underestimating R by less than $\frac{1}{n}$ means that the supremum of the integrals of all such f-underestimating simple functions will be $\geq R-\frac{1}{n}$. Since this is true for all n we must have that the lebesgue integral is R.

Step 3. To extend to arbitrary functions we simply use the positive part plus negative part characterisation of the Lebesgue integral and use linearity of the Riemann integral, then we're done.

□

Example. Consider the restriction of $1_{\mathbb{Q}}$ to $[0,1]$. Then we briefly saw in Level 8.1 that this is not Riemann integrable. However, it is Lebesgue integrable as it is the indicator function of a measurable set. In fact its integral is 0.

3.2 Basic properties of the Lebesgue integral

In this subsection we will prove the Monotone convergence theorem which is a very important tool, as well as important properties of the Lebesgue integral such as linearity.

Lemma. This is obvious. Let f,g be non-negative measurable. Then $f \leq g$ implies $\mu(f) \leq \mu(g)$ (Monotonicity of Lebesgue integral)

Proof. Everything $\leq f$ is $\leq g$ so the sup of simple functions $\leq f$ is less than the sup of simple functions $\leq g$

□

Theorem. (Monotone convergence theorem) Suppose that (f_n) is a sequence of non-negative measurable functions and $f_n \nearrow f$. Then f is non-negative measurable and $\mu(f_n) \nearrow \mu(f)$.

Proof. **Step 1.** f is non-negative measurable because limits of measurable functions are measurable by a theorem in the previous section.

Step 2. Suppose f_n and f are indicator functions, so that $f_n = 1_{A_n}$and $f = 1_A$. Then $f_n \nearrow f$ and therefore $A_n \nearrow A$.

Using the fact that if $A_n \nearrow A$ then $A = \cup A_n$ and so $\mu(A) = \lim \mu(A_n)$, ie continuity of measures, which follows from countable additivity, we note that $\mu(A_n) \nearrow \mu(A)$ and therefore $\mu(f_n) \nearrow \mu(f)$.

Step 3. Suppose f is an indicator function. Suppose $f = 1_A$ for some fixed A. Fix $\varepsilon > 0$ and set $A_n = \{f_n > 1-\varepsilon\}$ (which is in the σ-algebra by measurability of f_n). Then we know $A_n \nearrow A$ as every point in A_n eventually goes into A because $f_n \nearrow f$. Therefore we have

$$(1-\varepsilon)\, 1_{A_n} \leq f_n \leq f = 1_A$$

As $A_n \nearrow A$ we get

$$\lim_{n\to\infty} \mu(f_n) \geq (1-\varepsilon) \lim_{n\to\infty} \mu(A_n) = (1-\varepsilon)\, \mu\left(\lim_{n\to\infty} A_n\right) = (1-\varepsilon)\, \mu(f)$$

Since this is true for every ε the result follows.

Step 4. Suppose f is a simple function. We may suppose that each of the indicator functions that make up f are disjoint, as then we can apply the previous result after restricting f to each such indicator function and scaling it appropriately, and add up the finitely many results.

Step 5. Suppose f is arbitrary non-negative measurable. Then let $g \leq f$ be a simple function. Then we know from step 4 that $f_n \wedge g \nearrow f \wedge g = g$. Therefore we know that $\mu(f_n \wedge g) \to \mu(g)$

But then we have that $\lim(\mu(f_n)) \geq \lim(\mu(f_n \wedge g)) = \mu(g)$ Then since $\mu(g)$ can be made arbitrarily close to $\mu(f)$ by definition we have $\lim(\mu(f_n)) \geq \mu(f)$. But by monotonicity we have $\lim(\mu(f_n)) \leq \mu(f)$ so the limits are equal.

□

Theorem. We have the following properties

i) Let f, g be non-negative measurable functions between measure spaces. Then $\mu(af + bg) = a\mu(f) + b\mu(g)$ for $a, b \geq 0$ (Linearity of Lebesgue integral)

ii) Let f be non-negative measurable. Then clearly $\mu(f) = 0$ if and only if $f = 0$ almost everywhere (ie except on a measure 0 set).

Proof. i) Let $f_n = 2^{-n}\lfloor 2^n f \rfloor \wedge n$, ie each value of f is cut off after its nth binary place. Let g_n be defined analogously. Then f_n and g_n are simple with $f_n \nearrow f$ and $g_n \nearrow g$ and therefore $af_n + bg_n \nearrow af + bg$. Thus by the monotone convergence theorem we have $\mu(af_n + bg_n) \nearrow \mu(af + bg)$.

Since f_n and g_n are simple functions we have

$$\mu(af_n + bg_n) = a\mu(f_n) + b\mu(g_n) \nearrow a\mu(f) + b\mu(g)$$

Therefore $\mu(af_n + bg_n)$ tends to both $\mu(af + bg)$ and $a\mu(f) + b\mu(g)$ so they are equal.

ii) Suppose f is not 0 almost everywhere and let $A_n = \left\{x : f(x) > \frac{1}{n}\right\}$. Then their union is the set of points where $f > 0$. Since this has non-negative measure it follows that some A_n has negative measure. Suppose its measure is ε, then $\mu(f)$ must be at least $\frac{\varepsilon}{n}$.

Conversely, if f is 0 almost everywhere then you cannot make an indicator function that covers a non-zero measure set so you will have to ask for the supremum of stuff that are all 0 to get $\mu(f)$.

□

Corollary. If g_n is a sequence of non-negative measurable functions then $\mu\left(\sum g_n\right) = \sum(\mu(g_n))$

Theorem. Monotonicity and linearity hold for arbitrary measurable functions, not just non-negative ones. Furthermore for arbitrary measurable f, if it is 0 almost everywhere then its measure is 0.

Proof. The second part of the statement of the theorem is true with the same proof as in the previous theorem, therefore we just need to prove the first part.

By definition we have $\mu(-f) = -\mu(f)$. Given f, g let f^+ be $\max(f, 0)$ and f^- be $(-f)^+$ and let g^+ and g^- be defined similarly.

Suppose, then, that $a, b \geq 0$ since we can easily see by changing $g$to $-g$ if necessary that the result will follow from the restricted result to this assumption.

Also it is clear from the definition that $\mu(af) = a\mu(f)$. Therefore we just need to show that given f, g we have $\mu(f + g) = \mu(f) + \mu(g)$. We know this statement to be true for non-negative functions. Let $h = f + g$ and h^+ and h^- be defined analagously to f^+ and f^-. Then we have $h^+ - h^- = (f^+ - f^-) + (g^+ - g^-)$. We write

$$h^+ + f^- + g^- = h^- + f^+ + g^+$$

Now everything is non-negative measurable so we have linearity of μ here and can write

$$\mu\left(h^+\right) + \mu\left(f^-\right) + \mu\left(g^-\right) = \mu\left(h^-\right) + \mu\left(f^+\right) + \mu\left(g^+\right)$$

Rearranging gives

$$\mu\left(h^{+}\right)-\mu\left(h^{-}\right)=\mu\left(f^{+}\right)-\mu\left(f^{-}\right)+\mu\left(g^{+}\right)-\mu\left(g^{-}\right)$$

By definition of how we extend the measure to functions that are not necessarily non-negative the statement $\mu(h) = \mu(f) + \mu(g)$ follows so we are done.

Now we will prove monotonicity. Let $f \leq g$. Applying linearity and the fact that $\mu(g-f) \geq 0$, the result follows.

□

Proposition. Let A be a π-system with $E \in A$ and $\sigma(A) = \mathcal{E}$ and let f be an integrable function (with respect to a measure defined on $\mathcal{E}$) $E \to \mathbb{R}$ such that $\mu(f 1_a) = 0$ for all $a \in A$. Then $f = 0$ almost everywhere.

Proof. Let $D = \{a \in \mathcal{E} : \mu(f 1_a) = 0\}$. Then by properties of the integral this is a d-system (closedness under countable unions of increasing sets is by the monotone convergence theorem) so it equals $\mathcal{E}$ by Dynkin's lemma.

Now define $A^{+} = \{x \in E : f(x) > 0\}$ and $A^{-} = \{x \in E : f(x) < 0\}$. Then these are both in $\mathcal{E}$ by integrability of f so $\mu(f 1_{A^{+}}) = \mu(f 1_{A^{-}}) = 0$ so by the theorem about non-negative functions both $f 1_{A^{+}}$ and $f 1_{A^{-}}$ are 0 almost everywhere and thus so is f.

□

3.3 Limit theorems

Theorem. (Fatou's lemma) Let (f_n) be a sequence of non-negative measurable functions, then

$$\mu(\liminf(f_n)) \leq \liminf(\mu(f_n))$$

Proof. Note that trivially if $k \geq n$ then

$$\inf_{m \geq n} f_m \leq f_k$$

So by monotonicity we have

$$\mu\left(\inf_{m \geq n} f_m\right) \leq \mu(f_k)$$

For all $k \geq n$. Taking infimum on both sides we have

$$\mu\left(\inf_{m \geq n} f_m\right) \leq \inf_{k \geq n} \mu(f_k)$$

Keeping the definition of lim inf in mind, note that by the monotone convergence theorem the left hand side converges to $\mu(\liminf(f_n))$ and by the right hand side converges to $\liminf(\mu(f_n))$. Then the result follows as limits preserve non-strict inequalities.

□

Remark. To keep the direction correct in your mind suppose $f_n = 1_{[n,n+1]}$ so that $\mu(\liminf(f_n)) = \mu(0) = 0$ but $\liminf(\mu(f_n)) = 1$.

The next result is the dominated convergence theorem. We proved in level 6.2 a version of this in $\mathbb{R}^n$ that was extremely useful for fixing gaps in 'applied' maths, and now we will do it for more general measure spaces. Finally all the 'Borel' and 'σ-algebra' nonsense that I kept seeing when trying to make A level maths rigorous and was frustrated I didn't know what it meant, is actually something that I understand now and it actually isn't so big and scary once I know what's going on.

Theorem. (Dominated convergence theorem) Let f_n be a sequence of measurable functions such that $f_n \to f$ pointwise almost everywhere. Suppose that there is an integrable function g with $|f_n| \leq g$ for all n. Then f is integrable and $\mu(f_n) \to \mu(f)$.

Note that this essentially says you can swap limits and integrals under such conditions.

Proof. Note that since $|f_n| \leq g$ for all n and limits preserve non-strict inequalities we know that $|f| \leq g$ so $\mu(f) < \infty$. From this and the fact that limits of measurable functions are measurable, we know f is integrable.

Now we note that $g + f_n$ and $g - f_n$ are non-negative, measurable and integrable for all n.

We now will apply Fatou's lemma as follows:

$$\mu(f) = \mu(\liminf(f_n)) \leq \liminf(\mu(f_n))$$

We also gave that

$$\mu(g) - \mu(f) = \mu(g - f) = \mu(\liminf(g - f_n)) \leq \liminf(\mu(g - f_n))$$
$$\liminf(\mu(g) - \mu(f_n)) = \mu(g) + \liminf(-\mu(f_n)) = \mu(g) - \limsup(\mu(f_n))$$

Since $\mu(g)$ is finite we can subtract it from all the equations to get

$$\mu(f) \geq \limsup(\mu(f_n))$$

This is called the reverse Fatou lemma but it only necessarily holds when there is a dominating function g. These combine to tell us that

$$\mu(f) \geq \limsup(\mu(f_n)) \geq \liminf(\mu(f_n)) \geq \mu(f)$$

Thus these must all be equal to eachother, and so $\mu(f) = \lim(\mu(f_n))$

Note that since the convergence is almost everywhere, the resulting error from it not being everywhere is 0.

□

As an example from level 6.2 to show why the domination by g is necessary: If $f_n(x)$ is defined as n for $0<x<1/n$ then the integral of this will be 1, but at all points between 0 and 1 f_n will eventually go to 0, so the integral of the limit is not the limit of the integral. It turns out that in this case we cannot find a function g such that for all n we have $f_n < g$ and $\int_0^1 g < \infty$ as the integral of any "bounding" function will be infinite if we try to compute it, since we can bound it below by $\sum_{n=1}^{\infty} \frac{1}{n}$, which as we saw in the series solutions proof in level 6.1 grows like $\ln(n)$.

3.4 Constructing new measures (again)

We can restrict a measure space to a subset in the intuitive way.

Definition. (Restriction of measure space) Let $(E, \mathcal{E}, \mu)$ be a measure space and let $A \in \mathcal{E}$. Then the restriction of the measure space to A is $(A, \mathcal{E}_A, \mu_A)$ where $\mathcal{E}_A$ is the elements of $\mathcal{E}$ that are subsets of A and μ_A always agrees with μ. This gives a new measure space.

Lemma. Let $(E, \mathcal{E}, \mu)$ and $(F, \mathcal{F}, \mu')$ be measure spaces and let $A \in \mathcal{E}$. Let $f : E \to F$ be a measurable function. Then f restricted to A, which we write as $f|_A$, is $\mathcal{E}_A$-measurable.

Proof. We just need to show that pre-images of measurable sets are measurable. Let $B \in \mathcal{F}$ then we have

$$(f|_A)^{-1}(B) = f^{-1}(B) \cap A \in \mathcal{E}_A$$

□

We also have that we can integrate f over a measurable subset A of $\mathbb{R}$. In fact it is easy to see that the result is the same as if we just integrate $f1_A$ over $\mathbb{R}$.

We often write either $\int_E f1_A d\mu$ or $\int_A f d\mu$ and they mean the same thing.

Definition. Let $(E, \mathcal{E})$ and $(G, \mathcal{G})$ be measure spaces and let $f : E \to G$ be a measurable function. If μ is a measure on $E, \mathcal{E}$ then we can define $\nu = \mu \circ f^{-1}$ as the image measure or pushforward measure on $(G, \mathcal{G})$. Note that it is easy to check that it is countably additive and therefore a measure.

Proposition. Let $(E, \mathcal{E}, \mu)$ and $(G, \mathcal{G})$ be measure spaces and let $f : E \to G$ be measurable . Then let the measure ν on $(G, \mathcal{G})$ be defined by $\nu(A) = \mu\left(f^{-1}(A)\right)$ for all $A \in \mathcal{G}$. Then for any non-negative $\mathcal{G}$-measurable function $g : G \to [0, \infty]$ we have

$$\int_G g d\nu = \int_G (g \circ f) d\mu$$

Proof. Let $\mathcal{H}$ be the collection of all non-negative bounded $\mathcal{G}$-measurable functions $g : G \to \mathbb{R}$ that satisfy the proposition

$$\int_G g d\nu = \int_G (g \circ f) d\mu$$

or in other words

$$\nu(g) = \mu(g \circ f)$$

To use the monotone class theorem, we will let our π-system be $\mathcal{G}$ itself. To show that 1_A is in $\mathcal{H}$ for all $A \in \mathcal{G}$ we will prove the proposition for indicator functions. This is easy by our definitions:

$$\int_G 1_A d\nu = \nu(A) = \mu\left(f^{-1}(A)\right) = \int_E 1_{f^{-1}(A)} d\mu = \int_E 1_A \circ f d\mu$$

It then follows that 1_G is in $\mathcal{H}$. Next we need to show that $\mathcal{H}$ is a vector space. This follows by linearity of the integral - if 2 functions g and h satisfy the proposition so does $ag + bh$.

We now just need to check that $\mathcal{H}$ is closed under bounded, increasing limits. Let g_n be a sequence of non-negative functions in $\mathcal{H}$ such that $g_n \nearrow g$ pointwise where the limit function g is bounded. Since all g_n is in $\mathcal{H}$ we know that $\nu(g_n) = \mu(g_n \circ f)$ for all n. By the monotone convergence theorem on both measure spaces we have

$$\nu(g) = \lim(\nu(g_n)) = \lim(\mu(g_n \circ f)) = \mu(g \circ f)$$

Thus by the monotone class theorem $\mathcal{H}$ contains all bounded non-negative measurable functions.

To extend to unbounded functions g use the monotone convergence theorem on $g_n = \min(g, n)$.

$\square$

We can also define new measures using a *density*. This can be a probability density but it doesn't have to. Let $(E, \mathcal{E}, \mu)$ be a measure space and let f be a non-negative measurable function. then we can define $\nu(A) = \mu(f1_A)$ as a new measure on $(E, \mathcal{E})$

Proposition. ν as defined above is actually a measure.

Proof. Clearly it is 0 on the empty set. For countable additivity we note that if A_n is a disjoint sequence in $\mathcal{E}$ then

$$\nu(\cup A_n) = \nu(f1_{\cup A_n}) = \mu\left(f \sum 1_{A_n}\right) = \sum \mu(f1_{A_n}) = \sum \nu(A_n)$$

$\square$

Definition. We say a random variable X has a density f if its law μ_X (i.e. the corresponding probability measure) can be made this way using f as the density. In this case, we define

$$E[g(X)] = \int_{\mathbb{R}} g(x) f_X(x) dx$$

In the case f is Riemann integrable we get that it is the derivative of the cdf by the fundamental theorem of calculus. However f does not need to be unique as changing it on a measure 0 set gives a different valid density for the same random variable.

Example. (Cantor distribution) The following random variable cannot be made as the sum of discrete and continuous parts:

Construct the cdf as follows:

X is always between 0 and 1 so we make the cdf 0 or 1 outside of this range. Then for $\frac{1}{3} \leq X \leq \frac{2}{3}$ we set the cdf to $\frac{1}{2}$. We now repeat the following procedure recursively to every interval where we have not defined X:

Take each interval where X has not been defined and take the middle third. Then set the cdf to be halfway between what it was in the last interval we defined X before and the first interval we defined X after. For example, for $\frac{1}{9} \leq X \leq \frac{2}{9}$ we would set the cdf to $\frac{1}{4}$ and for $\frac{7}{9} \leq X \leq \frac{8}{9}$ we set the cdf to $\frac{3}{4}$. We then cover every interval this way which will cover almost all X values and a dense set of cdf values, and we can fill in the cdf everywhere else to make the function continuous. Then it is not possible to meaningfully assign a density at any point since at any point in one of our intervals. Note: It is useful to know that what is left when removing the interior of all these middle third intervals is called the Cantor Set, known for being closed, containing no isolated points, having measure 0 and being uncountable. The reason we cannot make this random variable part discrete part continuous is because for any point outside the cantor set the density would have to be 0 and at any point in the cantor set the cdf does not have a jump continuity so we can't assign a density or make it part discrete.

3.5 Integration and differentiation

We will prove a stronger version of integration by substitution and we will reprove differentiation under the integral sign.

Theorem. Let $\phi : [a, b] \to \mathbb{R}$ be continuously differentiable and increasing and let $g : [\phi(a), \phi(b)] \to \mathbb{R}$ be a bounded Borel measurable function. Then we have

$$\int_{\phi(a)}^{\phi(b)} g(y)\,dy = \int_a^b g(\phi(x))\,\phi'(x)\,dx$$

Proof. Let V be the set of borel functions g such that this holds. Then again it is a vector space by linearity of the integral. Of course, we want to use the monotone class theorem.

Consider the π-system of closed intervals $[u, v] \subseteq [\phi(a), \phi(b)]$. Then V contains the indicator functions of all these intervals by the fundamental theorem of calculus. This π-system generates the Borel σ-algebra on $[\phi(a), \phi(b)]$. To apply the monotone class theorem it remains to show that this is closed under bounded monotone limits. But this is again by the monotone convergence theorem: If we have

$$\int_{\phi(a)}^{\phi(b)} g_n(y)\,dy = \int_a^b g_n(\phi(x))\,\phi'(x)\,dx$$

and $g_n \nearrow g$ and g is bounded then g satisfies the formula. Therefore the conditions for the monotone class theorem are satisfied and this theorem is a direct corollary of the monotone class theorem.

$\square$

We will now show that under sufficient niceness conditions which we will specify we can say

$$\frac{d}{dt}\int f(t, x)\,dx = \int \frac{\partial}{\partial t} f(t, x)\,dx$$

Theorem. Let $(E, \mathcal{E}, \mu)$ be a measure space and let $U \subseteq \mathbb{R}$ be open and let $f : U \times E \to \mathbb{R}$ and suppose that f has the following properties

i) For any fixed $t \in U$ the map $x \longmapsto f(t, x)$ is integrable

ii) For any fixed $x \in E$ the map $t \longmapsto f(t,x)$ is differentiable

iii) There exists an integrable function $g : E \to \mathbb{R}_{\geq 0}$ such that for all $x \in E$ and $t \in U$ we have

$$\left| \frac{\partial}{\partial t} f(t,x) \right| \leq g(x)$$

Then the claim is that the map $x \longmapsto \frac{\partial f}{\partial t}(t,x)$ is integrable for all t and that the function $F(t) = \int_E f(t,x)\, d\mu$ is differentiable with derivative $\int_E \frac{\partial}{\partial t} f(t,x)\, d\mu$

Proof. Note that a derivative is measurable since it is the limit of measurable functions: Specifically if h_n is a non-zero sequence tending to 0 then the derivative is the limit of the measurable functions

$$\frac{f(t+h_n,x) - f(t,x)}{h_n}$$

Now define

$$g_n(x) = \frac{f(t+h_n,x) - f(t,x)}{h_n} - \frac{\partial}{\partial t} f(t,x)$$

The fact that such a sequence $t + h_n$ is in the domain is from the fact that U is open, which is where we use that fact.

Since f is differentiable we know that $g_n \to 0$ as $n \to \infty$. Also, by the mean value theorem we know that the first term of g_n is $\frac{\partial}{\partial t} f(t',x)$ for some t' so it is bounded above by $g(x)$. Therefore we have by the triangle inequality that

$$|g_n(x)| \leq 2g(x)$$

By definition of F we have

$$\frac{F(t+h_n,x) - F(t,x)}{h_n} - \int_E \frac{\partial}{\partial t} f(t,x)\, d\mu = \int_E g_n(x)\, d\mu$$

By dominated convergence we know the right hand side tends to 0 and therefore

$$\lim_{n\to\infty} \frac{F(t+h_n,x) - F(t,x)}{h_n} = \int_E \frac{\partial}{\partial t} f(t,x)\, d\mu$$

Since this is true for every sequence h_n that $\to 0$ it follows that F' exists and is equal to the integral.

□

4 Product measures

4.1 Definition and properties

Given measure spaces E_1 and E_2 we want to assign a measure to $E_1 \times E_2$. Define the product σ-algebra as follows:

Definition. Let $\mathcal{E}_1$ and $\mathcal{E}_2$ be σ-algebras on E_1 and E_2 respectively. Then define

$$\mathcal{A} = \{A_1 \times A_2 : A_1 \in E_1, A_2 \in E_2\}$$

This is a π-system on $E_1 \times E_2$ and the product σ-algebra is $\sigma(\mathcal{A})$ which we write as $\mathcal{E}_1 \otimes \mathcal{E}_2$.

Let μ_1 and μ_2 be measures on $\mathcal{E}_1$ and $\mathcal{E}_2$ respectively. We would like to define the measure of $A \in \mathcal{E}_1 \otimes \mathcal{E}_2$ as

$$\int_{E_1} \left(\int_{E_2} 1_A(x_1,x_2)\, d\mu_2 \right) d\mu_1$$

where $x_1 \in E_1$ and $x_2 \in E_2$. However, we need to check that this definition makes sense.

Lemma. Let $E = E_1 \times E_2$ be a measure space with the product σ-algebra $\mathcal{E}$ and let f be a $\mathcal{E}$-measurable function. Then

i) For each $x_1 \in E_1$ the function $x_2 \mapsto f(x_1, x_2)$ is $\mathcal{E}_1$-measurable

ii) If f is a non-negative measurable then $f_1(x_1) = \int_{E_2} f(x_1, x_2)\, d\mu_2$ is $\mathcal{E}_1$-measurable.

Proof. Note that for each fixed x_1 the map $\pi : E_2 \to E$ given by $\pi(x_2) = (x_1, x_2)$ is measurable (since the pre-image of anything in the generating set of $\mathcal{E}$ is $\mathcal{E}_2$-measurable) and since f is measurable and compositions of measurable maps are measurable the result of part (i) follows.

For part (ii) if we first assume f is bounded we can use the monotone class theorem. Let V be the set of all measurable functions such that $f_1(x_1) = \int_{E_2} f(x_1, x_2)\, d\mu_2$ is $\mathcal{E}_1$-measurable. Here note that we need to include ∞ in the codomain of f_1 since the integral is not necessarily finite. Then V is a vector space by linearity of the integral. It is also clear that if $A \in \mathcal{A}$ where $\mathcal{A}$ is as in the definition of the product σ-algebra then $1_A \in V$ and $1_E \in V$. Therefore we have the generating π-system condition and just need to check closure under monotone limits.

Let (f_n) be a non-negative sequence in V such that $f_n \nearrow f$ and f is bounded. Then by the monotone convergence theorem we have

$$\left(\int_{E_2} f_n(x_1, x_2)\, d\mu_2\right) \nearrow \left(\int_{E_2} f(x_1, x_2)\, d\mu_2\right)$$

Thus the function sending x_1 to $\int_{E_2} f(x_1, x_2)\, d\mu_2$ is a limit of measurable functions and therefore measurable. Therefore we conclude by the monotone class theorem.

If f is not bounded then by considering $f \wedge n$ and taking the limit in n we note that f is in V as well.

$\square$

Theorem. Suppose E_1 and E_2 are σ-finite measure spaces with measures μ_1 and μ_2 respectively on σ-algebras $\mathcal{E}_1$ and $\mathcal{E}_2$ and that $E_1 \times E_2 = E$ with the product σ-algebra. There exists a unique measurable function μ on $\mathcal{E}$ such that $\mu(A_1 \times A_2) = \mu(A_1)\, \mu(A_2)$ for all $A_1 \in \mathcal{E}_1, A_2 \in \mathcal{E}_2$

Proof. By σ-finiteness we can take a sequence of subsets $E_{1,n}$ and $E_{2,n}$ such that they are finite measure spaces and their unions are E_1 and E_2. We can then restrict to these subspaces $E_{1,n} \times E_{2,n}$, prove uniqueness there, and take a union/limit to get that uniqueness holds everywhere, since products preserve limits. Therefore, even if I've explained it terribly, we can reduce to the case where we assume E_1 and E_2 are finite.

We will now try to define $\mu(A)$ as

$$\int_{E_1} \left(\int_{E_2} 1_A(x_1, x_2)\, d\mu_2\right) d\mu_1$$

Note that this is well defined by the previous lemma, actually a measure since it is countably additive by the monotone convergence theorem and is 0 on the empty set, and we have a working measure satisfying the product relation.

Uniqueness follows by a theorem we proved at the beginning of this course that if measures agree on E and a generating π-system for the σ-algebra in question and E is finite then they are equal.

$\square$

Example. We can get non-uniqueness by taking the product of $[0, 1]$ with counting measure and $[0, 1]$ with Lebesgue measure.

Hint. Try making one measure by summing the integrals of a set A restricted to $y = k$ over all k. Try making another one by integrating the number of elements in the set A restricted to $x = t$ over all t. Show that these both satisfy the product condition but give different measures for the set $(x, x) : x \in [0, 1]$.

Remark. Actually by the same argument as for what we did in $\mathbb{R}$ with the whole Caratheodory extension theorem outer measure stuff, in $\mathbb{R}^2$ any subset of a measure 0 set is measurable. However, the set of $(0, x)$ for x in a non-measurable subset of $\mathbb{R}$ is such a set and is therefore measurable in $\mathbb{R}^2$ despite not living in $\mathcal{M} \otimes \mathcal{M}$. To allow us to use properties of product measure in $\mathbb{R}^n$, we will assume that we are indeed working in $\mathcal{M} \otimes \mathcal{M} ... \otimes \mathcal{M}$ even when we could in theory use a larger σ-algebra.

Note that we could have also defined the measure by

$$\int_{E_2} \left(\int_{E_1} 1_A (x_1, x_2)\, d\mu_1 \right) d\mu_2$$

But by uniqueness these are the same! We will show this in the following theorem.

Theorem. (Fubini/Tonelli's theorem) Let f be non-negative measurable on $E = E_1 \times E_2$ with respect to the product measure where E_1 and E_2 are both σ-finite measure spaces. Then

$$\int_{E_2} \left(\int_{E_1} f (x_1, x_2)\, d\mu_1 \right) d\mu_2 = \mu (f) = \int_{E_1} \left(\int_{E_2} f (x_1, x_2)\, d\mu_2 \right) d\mu_1$$

Proof. We will use the monotone class theorem. We know that this is true for all indicator functions in the σ-algebra on E. We know that the set V of functions f for which is true is a vector space by linearity of the integral. Of course, we want to use the monotone class theorem. We do the same old monotone convergence theorem for bounded and then $f \wedge n$ trick and then we know that V contains all non-negative measurable functions, so done.

□

Theorem. Let f be a Riemann integrable bounded function from a closed rectangle in $\mathbb{R}^n$ to $\mathbb{R}$. Then there exists a measurable function g such that $g = f$ almost everywhere and $\int_{\text{Lebesgue}} g = \int_{\text{Riemann}} f$

Proof. Let P_n be a partition of mesh 2^{-n} and let U_n and L_n be the best upper and lower step functions on P_n. Then $\int L_n \nearrow \int_{\text{Riemann}} f$ by Riemann integrability and therefore $\int_{\text{Riemann}} f = \int_{\text{Lebesgue}} \sup (L_n)$ by the monotone convergence theorem, as L_n is an increasing sequence and $\sup (L_n)$ is Borel measurable. We can do a similar argument to show that $\int_{\text{Riemann}} f = \int_{\text{Lebesgue}} \inf (U_n)$. Therefore $\int_{\text{Lebesgue}} (\inf (U_n) - \sup (L_n)) = 0$. Since $\inf (U_n) - \sup (L_n)$ is non-negative it must be 0 almost everywhere by a previous theorem about Lebesgue integrals. Thus $\sup (L_n)$ works as a g.

□

This is the best we can do since the indicator function of a non-measurable set in $\mathbb{R}$ restricted to a slice of $\mathbb{R}^2$ is Riemann integrable but not Lebesgue integrable with respect to the product measure. However the Lebesgue integral is generally still better since such functions are pathological anyways.

Proposition. Let E be as above and let f be integrable on E and let A be defined as

$$A = \left\{ x_1 \in E_1 : \int_{E_2} |f (x_1, x_2)|\, d\mu_2 < \infty \right\}$$

Then $\mu_1 (E_1 \setminus A) = 0$. Also, if we set

$$f_1 (x_1) = \begin{cases} \int_{E_2} f (x_1, x_2)\, d\mu_2 & x_1 \in A \\ 0 & \text{Otherwise} \end{cases}$$

then f_1 is a μ_1-integrable function and $\mu_1 (f_1) = \mu (f)$

Proof. Since the function $x_1 \longmapsto \int_{E_2} |f(x_1, x_2)|\, d\mu_2$ is μ_1-integrable by μ-integrability of f, it can only be infinite on a measure 0 set.

Now define

$$f_1^+(x_1) = \int_{E_2} [f(x_1, x_2) \vee 0]\, d\mu_2$$
$$f_1^-(x_1) = \int_{E_2} [f(x_1, x_2) \wedge 0]\, d\mu_2$$

Then by definition $f_1(x_1) = 1_{A_1}\left(f_1^+ - f_1^-\right)$ and the result follows since

$$\mu(f) = \mu\left(f^+\right) - \mu\left(f^-\right) = \mu_1\left(f_1^+\right) - \mu_1\left(f_1^-\right)$$

by Fubini/Tonelli. Therefore, since it is ok for functions to differ on a measure 0 set and it does not affect the integral it follows that

$$\mu(f) = \mu_1(f_1)$$

□

This means that as long as f is product measurable and the integral of its absolute value is finite, we can swap the order of integrals.

Theorem. Let $f : \mathbb{R} \to \mathbb{R}$ be non-negative measurable and integrable, then the area measure of the area under f equals $\mu(f)$

Proof. The proposition clearly holds for all simple functions and the set of functions for which it holds is clearly a vector space. We will know it is true for all bounded f if we can set up the monotone class theorem. Indeed it works by the usual monotone convergence theorem trick and the $\min(f, n)$ trick to extend to unbounded functions.

□

4.2 Application to probability

Proposition. Let $X_1, ..., X_n$ be random variables in $(\Omega, \mathcal{F}, P)$ with values in $(E_1, \mathcal{E}_1), ..., (E_n, \mathcal{E}_n)$ respectively. Then let $E = E_1 \times ... \times E_n$ with the product σ-algebra $\mathcal{E}$. Then $X = (X_1, X_2, ..., X_n)$ is $\mathcal{E}$-measurable

Proof. This follows as the generating set is just the sets of the form $A_1 \times A_2 \times ... \times A_n$ with $A_i \in \mathcal{E}_i$ for each i, and the pre-image of each of these generating sets is the intersection over all i from 1 to n of the pre-images of A_i, which is contained in $\mathcal{F}$ as $\mathcal{F}$ is a σ-algebra.

□

Proposition. Let $X_1, ..., X_n$ be as above. Then the following are equivalent:

i) They are independent

ii) $\mu_X = \bigotimes_{i=1}^n \mu_{X_i}$

iii) For all f bounded and measurable,

$$E\left[\prod_{k=1}^n f_k(X_k)\right] = \prod_{k=1}^n E\left[f_k(X_k)\right]$$

Proof. **Step 1.** (i) implies (ii)

Let $\nu = \bigotimes_{i=1}^{n} \mu_{X_i}$. Then we just need to check that ν and μ_X agree on the π-system generating the entire σ-algebra. Let

$$\mathcal{A} = \{A_1 \times ... \times A_n : \forall i A_i \in \mathcal{E}_i\}$$

Then for each $A \in \mathcal{A}$ we have, by independence,

$$\mu_X(A) = P[X \in A] = P\left[\bigcap X_i \in A_i\right] = \prod P[X_i \in A_i] = \nu(A)$$

Step 2. (ii) implies (iii)

We will use Fubini's theorem here. Note that

$$E\left[\prod_{k=1}^{n} f_k(X_k)\right] = \int_E \prod_{k=1}^{n} f_k(X_k)\, d\mu_X = \prod_{k=1}^{n} \int_{E_k} f_k(X_k)\, d\mu_{X_k} = \prod_{k=1}^{n} E[f_k(X_k)]$$

Step 3. (iii) implies (i)

Take $f_k = 1_{A_k}$ for $A_k \in \mathcal{E}_k$. Then we have

$$P\left[\bigcap X_i \in A_i\right] = E\left[\prod_{k=1}^{n} 1_{A_k}(X_k)\right] = \prod_{k=1}^{n} E[1_{A_k}(X_k)] = \prod_{k=1}^{n} P[X_i \in A_i]$$

Therefore the X_i's are independent.

$\square$

5 Inequalities and L^p spaces

5.1 Some definitions and inequalities

Definition. Let $(E, \mathcal{E}, \mu)$ be a measure space. For $1 \leq p < \infty$ define L^p to be the set of all measurable functions f such that

$$\|f\|_p := \left(\int |f|^p d\mu\right)^{1/p} < \infty$$

For $p = \infty$ we let L^∞ be the space of functions such that

$$\|f\|_\infty := \inf\{\lambda \geq 0 : |f| \leq \lambda \text{ almost everywhere}\} < \infty$$

To check that these are norms note that $\|f\|_p = 0$ only implies that $f = 0$ almost everywhere. We can solve this by identifying functions as the same if they differ on a set of measure 0.

To check that the triangle inequality holds for this norm we will need some inequalities.

Proposition. (Generalized Markov's inequality) Let f be non-negative measurable and let $\lambda > 0$. Then $\mu(\{f \geq \lambda\}) \leq \frac{\mu(f)}{\lambda}$

Proof.

$$f \geq f 1_{f \geq \lambda} \geq \lambda 1_{f \geq \lambda}$$

Taking μon every term gives the result.

$\square$

Proposition. Let $I \subseteq \mathbb{R}$ be an interval and let $c : I \to \mathbb{R}$ be convex. Let X be an integrable random variable with values in I. Then we have

$$E\left[c\left(X\right)\right] \geq c\left(E\left[X\right]\right)$$

Proof. The proof is the same as in IA probability. We note that c is measurable as convex functions equal the supremum of all lines lieing below them (with possibly different values at the endpoints if the interval is closed). By restricting to lines with rational slope and intercept we can make it a countable supremum.

□

Definition. Let $p, q \in [1, \infty]$. We say they are conjugate if $\frac{1}{p} + \frac{1}{q} = 1$. Note that we define $\frac{1}{\infty} = 0$.

Proposition. (Holder's inequality) Let $p, q \in [1, \infty]$ be conjugate. Then for f, g measurable, we have

$$\mu\left(|fg|\right) = \|fg\|_1 \leq \|f\|_p \|g\|_q$$

The special case when $p = q = 2$ is the Cauchy-Schwarz inequality which we proved in Level 7.3 (Vectors and matrices) for general vector spaces with an inner product.

Proof. First if p or q is infinite, say q is infinite, then

$$\|f\|_p \|g\|_q = \|f\|_1 \sup\left(|g|\right) = \int |f| \sup\left(|g|\right) \geq \int |fg|$$

Thus the inequality is proved for that case.

We will assume that both $\|f\|_p$ and $\|g\|_q$ are finite as otherwise the proposition is trivial. We will scale appropriately so that both norms are 1 and then we just need to show that $\int |fg| \, d\mu \leq 1$

Now since $\frac{1}{p} + \frac{1}{q} = 1$ we will let X be a random variable that takes a with probability $\frac{1}{p}$ and b with probability $\frac{1}{q}$ and suppose $a, b > 0$ and consider the function $-\log(x)$ on $(0, \infty)$. Then this function is convex so Jensen's inequality tells us that

$$-\frac{1}{p}\log\left(a\right) - \frac{1}{q}\log\left(b\right) \geq -\log\left(\frac{a}{p} + \frac{b}{q}\right)$$

Therefore

$$\frac{1}{p}\log\left(a\right) + \frac{1}{q}\log\left(b\right) \leq \log\left(\frac{a}{p} + \frac{b}{q}\right)$$

Also, by renaming a to a^p and b to b^q we have

$$\frac{1}{p}\log\left(a^p\right) + \frac{1}{q}\log\left(b^q\right) \leq \log\left(\frac{a^p}{p} + \frac{b^q}{q}\right)$$

Exponentiating both sides gives

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

Therefore we have, by monotonicity of the integral

$$\int |fg| \, d\mu \leq \int \frac{|f|^p}{p} + \frac{|g|^q}{q} d\mu = \frac{1}{p} + \frac{1}{q} = 1$$

□

We will use this to prove Minkowski's inequality which will be useful. It is essentially the triangle inequality which establishes that the L^p norms are actually norms.

Lemma. Let $a, b \geq 0$ and let $p \geq 1$. Then $(a+b)^p \leq 2^p (a^p + b^p)$. Note that this is a somewhat trivial bound but it is useful anyway.

Proof. Suppose $a \leq b$ as the same argument works if $b \leq a$. Then

$$(a+b)^p \leq (2b)^p = 2^p b^p \leq 2^p (a^p + b^p)$$

$\square$

Theorem. (Minkowski inequality) Let $p \in [1, \infty]$ and f, g measurable. Then

$$\|f+g\|_p \leq \|f\|_p + \|g\|_p$$

Proof. If $p = 1$ or $p = \infty$ this just follows from the normal triangle inequality. If $\|f+g\|_p = 0$ then the proposition is trivially true. If $\|f+g\|_p = \infty$ then the right hand side is also infinite because we have

$$\infty = \int |f+g|^p \leq \int (|f| + |g|)^p \leq 2^p \int |f|^p + |g|^p$$

So now suppose that the left hand side is finite and $p \in (1, \infty)$. Then we have

$$\mu(|f+g|^p) = \mu\left(|f+g|\,|f+g|^{p-1}\right) \leq \mu\left(|f|\,|f+g|^{p-1}\right) + \mu\left(|g|\,|f+g|^{p-1}\right)$$

Now let q be conjugate to p. Then by the previous inequality we have

$$\leq \|f\|_p \left\||f+g|^{p-1}\right\|_q + \|g\|_p \left\||f+g|^{p-1}\right\|_q = \left(\|f\|_p + \|g\|_p\right) \left\||f+g|^{p-1}\right\|_q$$

$$= \left(\|f\|_p + \|g\|_p\right) \mu\left(|f+g|^{q(p-1)}\right)^{\frac{1}{q}} = \left(\|f\|_p + \|g\|_p\right) \mu\left(|f+g|^{\left(\frac{1}{1-\frac{1}{p}}\right)(p-1)}\right)^{1-\frac{1}{p}}$$

$$= \left(\|f\|_p + \|g\|_p\right) \mu(|f+g|^p)^{1-\frac{1}{p}}$$

Therefore we know that

$$\mu(|f+g|^p) \leq \left(\|f\|_p + \|g\|_p\right) \mu(|f+g|^p)^{1-\frac{1}{p}}$$

Therefore dividing both sides by $\mu(|f+g|^p)^{1-\frac{1}{p}}$ tells us

$$\mu(|f+g|^p)^{\frac{1}{p}} \leq \left(\|f\|_p + \|g\|_p\right)$$

which is equivalent to the statement of the theorem.

$\square$

5.2 L^p spaces

As a reminder the norm of a vector space is a function $\|\bullet\| : V \to \mathbb{R}_{\geq 0}$ such that it satisfies the triangle inequality, $(\|v\| = 0) \iff v = 0$ and the fact that for all $\alpha \in \mathbb{R}$ we have $\|\alpha v\| = |\alpha| \, \|v\|$

Now given a measure space $(E, \mathcal{E}, \mu)$ we define the L^p vector space as the set of measurable functions with finite L^p norm, as we saw at the start of the last section, except now we know that we actually have a norm, provided we identify elements that are the same almost everywhere. Specifically, we define equivalence classes as follows:

$$[f] = \{g \in L^p : f - g = 0 \text{ a.e.}\}$$

and we make our vector space be on the equivalence classes.

Recall that a metric space is complete if every Cauchy sequence converges.

Definition. Let V be a vector space with a norm, and define a metric on V by $d(v_1, v_2) = \|v_1 - v_2\|$. Then if this is a complete metric space, we say V is a Banach space.

Theorem. Let $1 \leq p \leq \infty$, then L^p is a Banach space.

Proof. If $p = \infty$ this follows from completeness of $\mathbb{R}$ as a metric space, and the fact that if we have a Cauchy sequence of functions under the infinity norm then their pointwise limit is a limit to the Cauchy sequence and has finite infinity norm and limits of measurable functions are measurable. Therefore we only need to prove it for $p < \infty$.

Let (f_n) be a sequence in L^p with $\|f_n - f_m\|_p \to 0$ as $(n, m) \to \infty$. Then there exists a subsequence (f_{n_k}) of f_n such that for all $k \in \mathbb{N}$ we have

$$\left\|f_{n_{k+1}} - f_{n_k}\right\|_p \leq 2^{-k}$$

Then for all M by Minkowski's inequality applied to the functions $f_{n_{k+1}} - f_{n_k}$ and the fact that it is always true that $\|f\|_p = \||f|\|_p$ we have

$$\left\|\sum_{k=1}^{M} \left|f_{n_{k+1}} - f_{n_k}\right|\right\|_p \leq \sum_{k=1}^{M} \left\|f_{n_{k+1}} - f_{n_k}\right\|_p \leq 1 \tag{1}$$

We know that

$$\sum_{k=1}^{M} \left|f_{n_{k+1}} - f_{n_k}\right| \nearrow \sum_{k=1}^{\infty} \left|f_{n_{k+1}} - f_{n_k}\right|$$

by definition of an infinite sum. By the monotone convergence theorem it then follows that we can take the limit on both sides of the equation labelled (1) and we get that

$$\left\|\sum_{k=1}^{\infty} \left|f_{n_{k+1}} - f_{n_k}\right|\right\|_p \leq \sum_{k=1}^{\infty} \left\|f_{n_{k+1}} - f_{n_k}\right\|_p \leq 1$$

Therefore we must have that $\sum_{k=1}^{\infty} \left|f_{n_{k+1}} - f_{n_k}\right| < \infty$ almost everywhere. Therefore, f_{n_k} converges almost everywhere, since the sequence of partial sums of $f_{n_{k+1}} - f_{n_k}$ converges almost everywhere as it is absolutely convergent almost everywhere. Then we set f to be the limit of this sequence at all these almost everywhere points and 0 everywhere else.

This is measurable as it is a limit of measurable functions almost everywhere and we just change it on a measure 0 set. In $\mathbb{R}^n$, if the measure 0 set happens to be non-measurable, pick a measurable measure 0 set it is contained in and make f 0 there.

Now to prove the theorem we just need to check that f is actually a limit of the Cauchy sequence and that $\|f\|_p < \infty$. To check it is a Cauchy sequence we use Fatou's lemma:

$$\|f_n - f\|_p = \mu\left(|f_n - f|^p\right)^{\frac{1}{p}} = \mu\left(\liminf_{k \to \infty} |f_n - f_{n_k}|^p\right)^{\frac{1}{p}}$$

The last equality is because the 2 functions are equal almost everywhere so they have the same integral. We use Fatou's lemma to get

$$\|f_n - f\|_p \leq \liminf_{k \to \infty} \mu\left(|f_n - f_{n_k}|^p\right)^{\frac{1}{p}}$$

Since the sequence is Cauchy the RHS tends to 0 and thus the LHS does too so the sequence converges. Now we just need to check that $\|f\|_p < \infty$ to see that $f \in L^p$. We have

$$\mu\left(|f|^p\right) = \mu\left(|f - f_n + f_n|^p\right) \leq \mu\left((|f - f_n| + |f_n|)^p\right) \leq 2^p \mu\left(|f - f_n|^p + |f_n|^p\right) = 2^p \left(\mu\left(|f - f_n|^p\right) + \mu\left(|f_n|^p\right)\right)$$

The second term is always finite and if we pick n large enough the first term will always be finite if it tends to 0. Therefore there exists an n that makes the last term finite, thus the first term, so $f \in L^p$. So done.

□

5.3 The L^2 space

By Holder's inequality, if f, g are in L^2 then fg is guaranteed to be integrable, and this argument only works when $p = 2$. This therefore means that we can sensibly define an inner product on the L^2 space by

$$\langle f, g \rangle = \int (fg)\, d\mu$$

Then this induces the L^2 norm by

$$\|f\|_2^2 = \langle f, f \rangle$$

Definition. A vector space is a Hilbert space if it has a complete inner product (recall that this is a symmetric bilinear positive definite function for real vector spaces and a conjugate-symmetric Hermitian positive definite function for complex vector spaces).

Definition. Two functions $f, g \in L^2$ are said to be orthogonal if $\langle f, g \rangle = 0$.

Definition. Let $V \subseteq L^2$. Then we define

$$V^\perp = \left\{f \in L^2 : \langle f, v \rangle = 0 \text{ for all } v \in V\right\}$$

We call this the orthogonal complement of V.

Definition. Let $V \subseteq L^2$. Then V is said to be closed if for every sequence in V its limit is in the same equivalence class as an element of V.

What makes L^2 nice is the following theorem:

Theorem. Let V be a closed subspace of L^2. Then each $f \in L^2$ can be decomposed as $f = u + v$ for $v \in V$ and $u \in V^\perp$. Furthermore for every $g \in V$ we have

$$\|f - v\|_2 \leq \|f - g\|_2$$

for all $g \in V$ with equality if and only if $g \sim v$. (Note: $\sim$ means in the same equivalence class, i.e. same except in a measure 0 set)

Proof. To prove this we will use the following 2 facts which follow by linearity with the same proof as in $\mathbb{R}^n$.

$$\text{Pythagoras identity } \|f + g\|^2 = \|f\|^2 + \|g\|^2 + 2\langle f, g \rangle$$

$$\text{Parallelogram law } \|f + g\|^2 - \|f - g\|^2 = 2\left(\|f\|^2 + \|g\|^2\right)$$

Given $f \in L^2$ we will take a sequence (g_n) in V such that

$$\|f - g_n\|_2 \to \inf_{g \in V} \|f - g\|_2 =: d(f, V)$$

We will now show that g_n is a Cauchy sequence.

Note: from now on in this proof if I say $\|\cdot\|$ I mean $\|\cdot\|_2$

Let n, m be natural numbers. Then by the parallelogram law with $f - g_n$ and $f - g_m$ we know

$$\|2f - g_n - g_m\|^2 + \|g_m - g_n\|^2 = 2\left(\|f - g_n\|^2 + \|f - g_m\|^2\right)$$

Therefore

$$\|g_m - g_n\|^2 = 2\left(\|f - g_n\|^2 + \|f - g_m\|^2\right) - 4\left\|f - \frac{g_n + g_m}{2}\right\|^2$$

As $n, m \to \infty$ the $2\left(\|f - g_n\|^2 + \|f - g_m\|^2\right)$ term tends to $4d(f, V)^2$ and by definition of $d(f, V)$ the last term is bounded below in magnitude by $4d(f, V)^2$. Therefore as $n, m \to \infty$ we have that the right hand side $\to 0$ so that means that g_n is a Cauchy sequence so it has a limit v which is contained in V by closedness.

By the dominated convergence theorem we can swap limits and norms which means that

$$\|f - v\| = \lim \|f - g_n\| = d(f, V)$$

Thus the infimum $\inf_{g \in V} \|f - g\|_2$ is attained by some $v \in V$. We now want to show that $u := f - v$ is in $V^\perp$ to complete the decomposition. This amounts to showing that $\langle u, h\rangle = 0$ for every $h \in V$.

Now let $t \in \mathbb{R}$. Then we have by definition

$$d(f, V)^2 \leq \|f - (v + th)\|_2^2 = \|f - v\|^2 + t^2\|h\|^2 + 2t\langle (f - v), h\rangle$$

This is a quadratic in t that is minimized when $t = 0$ by definition of $d(f, V)$ but on the other hand it is minimized when $t = \frac{\langle (f-v), h\rangle}{\|h\|^2}$. This means that $\langle (f - v), h\rangle = 0$ as required.

Now the inequality shown in the statement of the theorem is immediate from our work. To prove the equality condition note that if $f = u + v$ then $f - g = u + (v - g)$ and $u, v - g$ are orthogonal. Then note that by the pythagorean identity we have

$$\|f - g\|^2 = \|u\|^2 + \|v - g\|^2 = \|f - v\|^2 + \|v - g\|^2$$

Therefore if we do not have the $v \sim g$ condition we do not have equality.

$\square$

5.4 Conditional expectation

Definition. Suppose we have a probability space $(\Omega, \mathcal{F}, P)$ and let G_n be a collection of pairwise disjoint events with $\cup_n G_n = \Omega$ and $P(G_n) \neq 0$ for all n. Then let

$$\mathcal{G} = \sigma(G_n : n \in \mathbb{N})$$

i.e. the smallest σ-algebra containing all information about the G_n's. Then we define

$$E[X|G_n] = \frac{E[X 1_{G_n}]}{P[G_n]}$$

We can also define the conditional expectation of X given $\mathcal{G}$ as the random variable

$$Y = \sum_{n=1}^{\infty} E\left[X|G_n\right] 1_{G_n}$$

This, intuitively, rolls an outcome from Ω, checks which G_n it is in, then returns the expectation of another random variable X given that we are in that G_n.

Note that if $X \in L^2(P)$ (this is actually just the space of variables with finite variance) then $Y \in L^2(P)$ and clearly $Y$is $\mathcal{G}$-measurable.

Proposition. The conditional expectation of X given $\mathcal{G}$ is the projection of X onto the subspace $L^2(\mathcal{G}, P)$ of $\mathcal{G}$-measurable L^2 random variables in the space $L^2(P)$

Proof. Note that $L^2(\mathcal{G}, P)$ is actually a closed subspace as $\mathcal{G}$-measurable functions are closed under limits and so are L^2 spaces themselves.

It now suffices to show, by previous work, that $E\left[(X - W)^2\right]$ is minimized among $\mathcal{G}$-measurable random variables when $W = Y$.

If W is $\mathcal{G}$-measurable it follows that we can write

$$W = \sum_{n=1}^{\infty} a_n 1_{G_n}$$

where $a_n \in \mathbb{R}$. Then

$$E\left[(X - W)^2\right] = E\left[\left(\sum_{n=1}^{\infty} (X - a_n) 1_{G_n}\right)^2\right]$$

$$= E\left[\sum_{n=1}^{\infty} \left(X^2 + a_n^2 - 2a_n X\right) 1_{G_n}\right] = E\left[\sum_{n=1}^{\infty} \left(X^2 + a_n^2 - 2a_n E\left[X|G_n\right]\right) 1_{G_n}\right]$$

The last equality comes from the fact that

$$E\left[X 1_{G_n}\right] = E\left[X|G_n\right] P\left[G_n\right] = E\left[X|G_n\right] E\left[1_{G_n}\right] = E\left[E\left[X|G_n\right] 1_{G_n}\right]$$

Now the quadratic in the expectation over a_n is minimized for $a_n = E\left[X|G_n\right]$ which corresponds to $W = Y$.

$\square$

Note that we can interpret

$$\operatorname{Var}\left[X\right] = \|X - E\left[X\right]\|_2^2$$
$$\operatorname{Cov}\left[X, Y\right] = \langle X - E\left[X\right], Y - E\left[Y\right]\rangle$$

5.5 Convergence in $L^1(P)$

Convergence in $L_1(P)$ means that for all X, $E\left[X_n - X\right] \to 0$ as $n \to \infty$. However, convergence in probability and almost sure convergence both do not imply this.

Example. Start by rolling a random variable Y from $U\left[0, 1\right]$ and then let $X_n = \begin{cases} n & \text{Y} < \frac{1}{n} \\ 0 & Y \geq \frac{1}{n} \end{cases}$. Formally, let

$$(\Omega, \mathcal{F}, P) = ((0, 1), \mathcal{B}(0, 1), \text{Lebesgue})$$

and let X_n be the function $n1_{(0,\frac{1}{n})}$

in fact every random variable can in theory be expressed as a function from $(0,1)$ like this

Then X_n converges to 0 almost surely, therefore in probability, but $E\left[|X_n - X|\right] \to 1$.

However, there is a theorem that prevents this kind of thing from happening.

Theorem. (Bounded convergence theorem) Let X, X_n be random variables such that $X_n \to X$ in probability and suppose there exists a constant C such that $|X_n| \leq C$ for all n. Then $X_n \to X$ in L^1.

Proof. Let $\varepsilon > 0$ be arbitrary. Then

$$P\left[|X| > C + \varepsilon\right] \leq P\left[|X - X_n| + |X_n| > C + \varepsilon\right] \leq P\left[|X - X_n| > \varepsilon\right] + P\left[|X_n| > C\right]$$

Then as $n \to \infty$ this $\to 0$ and therefore

$$P\left[|X| > C + \varepsilon\right] = 0$$

Since this is true for every ε we have that $|X| \leq C$ almost everywhere.

Now fix another $\varepsilon > 0$. Then

$$E\left[|X_n - X|\right] = E\left[|X_n - X|\, 1_{|X_n - X| \leq \frac{\varepsilon}{2}} + |X_n - X|\, 1_{|X_n - X| > \frac{\varepsilon}{2}}\right] \leq \frac{\varepsilon}{2} + 2CP\left[|X - X_n| > \varepsilon\right]$$

Since the last term is $< \frac{\varepsilon}{2}$ for large enough n by convergence in probability we have that for large enough n,

$$E\left[|X_n - X|\right] < \varepsilon$$

Therefore are done since ε was arbitrary.

□

5.6 Uniform integrability

Definition. Let $\mathcal{X}$ be a family of random variables (as defined by Borel functions from $(0,1) \to \mathbb{R}$). Then $\mathcal{X}$ is said to be L^p-bounded if

$$\sup_{X \in \mathcal{X}} \left\{ \|X\|_p : X \in \mathcal{X} \right\} < \infty$$

Definition. Let $\mathcal{X}$ be a family of random variables and define

$$I_{\mathcal{X}}(\delta) := \sup_{A \in \mathcal{F}} \left\{ E\left[|X|\, 1_A\right] : X \in \mathcal{X}, A \in \mathcal{F}, P[A] < \delta \right\}$$

Then $\mathcal{X}$ is said to be uniformly integrable if $\mathcal{X}$ is L^1-bounded and $I_X(\delta) \to 0$ as $\delta \to 0$

This is sometimes called being Aboslutely Continuous.

Note that clearly, finite unions of uniformly integrable sets are uniformly integrable. Also, we have the following:

Proposition. Let $\mathcal{X}$ be an L^p-bounded family for some $p > 1$, then $\mathcal{X}$ is uniformly integrable.

Proof. Note that if $p = \infty$ then this corresponds to an almost surely bounded random variable so clearly the family is also, say L^2 bounded if it is L^∞ bounded. Therefore assume p is finite.

Now let
$$C = \sup_{X \in \mathcal{X}} \left\{ \|X\|_p : X \in \mathcal{X} \right\} < \infty$$

this is possible by L^p-boundedness.

Now let $X \in \mathcal{X}$ and $A \in \mathcal{F}$. Then let $p$and $q$be conjugate, then by Holder's inequality we have

$$E\left[|X| 1_A\right] \leq \|X\|_p P\left[A\right]^{\frac{1}{q}} \leq CP\left[A\right]^{\frac{1}{q}}$$

This is now a uniform bound depending only on C and $P[A]$ that goes to 0 as $P[A]$ goes to 0 so we have uniform integrability.

$\square$

Note that if $p = 1$ the above theorem fails by our earlier example.

Proposition. Let $\mathcal{X}$ be a family of random variables. Then $\mathcal{X}$ is uniformly integrable if and only if as $k \to \infty$

$$\sup_{X \in \mathcal{X}} \left\{ E\left[|X| 1_{|X|>k}\right] \right\} \to 0$$

Proof. First suppose $\mathcal{X}$ is uniformly integrable. Then for any k we have, by Markov's inequality

$$P\left[|X| \geq k\right] \leq \frac{E\left[|X|\right]}{k}$$

By uniform integrability, given $\varepsilon > 0$ we can find δ such that $P\left[|X| 1_A\right] < \varepsilon$ for all A with $\mu(A) < \delta$. Then pick k sufficiently large such that
$$k\delta > \sup_{X \in \mathcal{X}} \left\{ E\left[X\right] : X \in \mathcal{X} \right\}$$
Possible because uniform integrability requires L^1-boundedness. Then $P\left[|X| \geq k\right] < \delta$ and therefore $E\left[|X| 1_{|X|>k}\right] < \varepsilon$ for all $X \in \mathcal{X}$.

Conversely suppose that as $k \to \infty$
$$\sup_{X \in \mathcal{X}} \left\{ E\left[|X| 1_{|X|>k}\right] \right\} \to 0$$

Then we need to show that $\mathcal{X}$ is L^1-bounded and satisfies the other condition. For L^1-boundedness we have

$$E\left[|X|\right] = E\left[|X| \left(1_{|X|\leq k} + 1_{|X|>k}\right)\right] \leq k + E\left[|X| \left(1_{|X|>k}\right)\right]$$

The term $E\left[|X| \left(1_{|X|>k}\right)\right]$ is bounded by, say, 1 for sufficiently large k which implies $E\left[|X|\right]$ is bounded by $k+1$, so we have L^1-boundedness.

Also for any measurable A and $X \in \mathcal{X}$ we have

$$E\left[|X| 1_A\right] = E\left[|X| 1_A \left(1_{|X|\leq k} + 1_{|X|>k}\right)\right] \leq E\left[|X| 1_{|X|>k}\right] + kP\left[A\right]$$

For any $\varepsilon > 0$ we can pick k sufficiently large so that the first term is $< \frac{\varepsilon}{2}$ for all $X \in \mathcal{X}$ by assumption. Then whenever $P\left[A\right] < \frac{\varepsilon}{2k}$ we have $E\left[|X| 1_A\right] < \varepsilon$ which implies uniform integrability.

$\square$

Corollary. A finite collection of L^1 functions is uniformly integrable.

Proof. Note that finite unions of uniformly integrable sets are uniformly integrable so we just need to check that a set containing a single L^1 function is uniformly integrable. To do this note that

$$E\left[|X|\right] = \sum_{k=0}^{\infty} E\left[|X| \, 1_{|X| \in [k,k+1)}\right]$$

Since the sum is a finite value we must have that as $k \to \infty$

$$E\left[|X| \, 1_{|X| \geq k}\right] \to 0$$

thus X is uniformly integrable by the previous proposition.

$\square$

Theorem. Let X be a random variable and let (X_n) be a sequence of random variables. Suppose $X_n, X \in L^1$ for all n and $X_n \to X$ in L^1. Then $\{X_n\}$ is uniformly integrable and $X_n \to X$ in probability.

Proof. **Step 1.** Convergence in probability

Let $\varepsilon > 0$. Then by Markov's inequality we have

$$P\left(|X_n - X| \geq \varepsilon\right) \leq \frac{E\left[|X_n - X|\right]}{\varepsilon}$$

Since $X_n \to X$ in L^1 the numerator tends to 0 and therefore $P\left(|X_n - X| \geq \varepsilon\right) \to 0$ and this is true for every ε.

Step 2. Uniform integrability

For each X_n we have by the triangle inequality

$$E\left[|X_n|\right] = E\left[|X_n - X + X|\right] \leq E\left[|X_n - X|\right] + E\left[|X|\right]$$

Since $E\left[|X_n - X|\right]$ is finite for all n and tends to 0 it is a bounded sequence so $E\left[|X_n|\right]$ is bounded so we have L^1-boundedness.

To get the other condition for uniform integrability we let $\varepsilon > 0$. By convergence in L^1 there is an N such that $n > N$ implies

$$E\left[|X_n - X|\right] < \frac{\varepsilon}{2}$$

For any arbitrary event A we have

$$E\left[|X_n| \, 1_A\right] \leq E\left[|X_n - X| \, 1_A\right] + E\left[|X| \, 1_A\right] \leq E\left[|X_n - X|\right] + E\left[|X| \, 1_A\right] < \frac{\varepsilon}{2} + E\left[|X| \, 1_A\right]$$

Since $X \in L^1$ and it is a single random variable it is uniformly integrable, which means that there exists a $\delta_0 > 0$ such that if $P(A) < \delta_0$ then $E\left[|X| \, 1_A\right] < \frac{\varepsilon}{2}$. Therefore, as $P[A] \to 0$ we have $E\left[|X_n| \, 1_A\right] \to 0$ uniformly in n so uniform integrability holds.

$\square$

Theorem. The converse to the above theorem is true. Let X be a random variable and let (X_n) be a sequence of random variables. Suppose $\{X_n\}$ is uniformly integrable and $X_n \to X$ in probability. Then $X_n, X \in L^1$ for all n and $X_n \to X$ in L^1.

Proof. X_n is uniformly integrable so all X_n are in L^1. However, we need to prove that $X \in L^1$.

Recall that we previously showed that if we have sequence of random variables which converge in probability, we can exhibit a subsequence which converges almost surely. Let X_{n_k}be such a sequence in this case. Using this fact, and Fatou's lemma, and L^1-boundedness in that order, we have

$$E[|X|] = E\left[\liminf_{k\to\infty} |X_{n_k}|\right] \leq \liminf_{k\to\infty} E[|X_{n_k}|] < \infty$$

This establishes that $X \in L^1$. We now want to show that $X_n \to X$ in L^1.

Fix $\varepsilon > 0$. Then there exists $K \in (0,\infty)$ such that we have both of

$$E\left[|X| 1_{\{|X|>K\}}\right] \leq \frac{\varepsilon}{3}$$
$$E\left[|X_n| 1_{\{|X_n|>K\}}\right] \leq \frac{\varepsilon}{3}$$

This is by out characterisation of uniform integrability. Now define

$$X_n^K = (X_n \vee -K) \wedge K$$
$$X^K = (X \vee -K) \wedge K$$

This is essentially just X_n and X but capped to always be between -K and K so that intuitively if we roll an instance of the random variable and it is outside that range we set it to -K or K accordingly.

Note that since $X_n \to X$ in probability, clearly $X_n^K \to X^K$ in probability. This is since we will always have $\left|X_n^K - X^K\right| \leq |X_n - X|$.

Now the bounded convergence theorem tells us that there is some Nsuch that if $n \geq N$ then

$$E\left[\left|X_n^K - X^K\right|\right] \leq \frac{\varepsilon}{3}$$

Now for all $n \geq N$ we have

$$E[|X_n - X|] \leq E\left[\left|X_n - X_n^K\right| + \left|X_n^K - X^K\right| + \left|X^K - X\right|\right]$$
$$\leq E\left[\left|X_n^K - X^K\right|\right] + E\left[|X| 1_{\{|X|>K\}}\right] + E\left[|X_n| 1_{\{|X_n|>K\}}\right] \leq \varepsilon$$

This establishes convergence in L^1.

□

6 The Fourier transform

6.1 Definition and basic properties

When talking about Fourier transforms we care about functions from $\mathbb{R}^d \to \mathbb{C}$. Therefore, we will write L^p for complex-valued Borel functions on $\mathbb{R}^d$ with $\|f\|_p = \left(\int_{\mathbb{R}^d} |f|^p\right)^{\frac{1}{p}} < \infty$.

Note that these integrals of the real and imaginary parts are defined because taking the real or imaginary part is a measurable function and we would just be composing those, and by monotonicity the integral of the separate parts will be smaller than the integral of the whole thing (after taking absolute values) and thus finite.

Definition. Let $f \in L^1\left(\mathbb{R}^d\right)$. Then its Fourier transform $\hat{f} : \mathbb{R}^d \to \mathbb{C}$ is given by

$$\hat{f}(u) = \int_{\mathbb{R}^d} f(x)\, e^{i(u.x)} dx$$

We will see throughout level 11 (particularly in the actual Methods course) why Fourier transforms are useful.[1]

Definition. Let μ be a finite measure on $\mathbb{R}^d$ then we define its Fourier Transform to be a function $\mu : \mathbb{R}^d \to \mathbb{C}$ given by

$$\hat{\mu}(u) = \int_{\mathbb{R}^d} e^{i(u.x)} d\mu$$

If X is a random variable with corresponding law μ_X then the characteristic function of X, which we will have defined in levels 6.3 and 8.4, will be given by

$$\phi_X(u) = \hat{\mu}_X(u) = E\left[e^{i(u.x)}\right]$$

We note that $\left\|\hat{f}\right\|_\infty \leq \|f\|_1$ by the triangle inequality for integrals and that $\|\hat{\mu}\|_\infty \leq \mu\left(\mathbb{R}^d\right)$.

Proposition. The functions $\hat{f}$ and $\hat{\mu}$ are continuous

Proof. In the case of $\hat{f}$ it is because $f(x) e^{i(u_n.x)}$ approaches $f(x) e^{i(u..x)}$ for any sequence $u_n \to u$ pointwise and $|f|$ is a dominating function in L^1.

For $\hat{\mu}$ the argument is the same if we use the fact that we assume the measure is finite when we make $\hat{\mu}$

□

6.2 Convolutions

Definition. Let μ, ν be probability measures that are the law of random variables X and Y respectively. Then their convolution, denoted $\mu * \nu$, is the law of $X + Y$. Specifically,

$$\mu * \nu(A) = P[X + Y \in A] = \iint 1_A(x + y)\, d\mu d\nu$$

Now suppose that X has density f. Then we have (Fubini's theorem allowing swapping of integrals here since the integrand is non-negative)

$$\mu * \nu(A) = \iint 1_A(x + y) f(x)\, dx d\nu$$

$$= \iint 1_A(x) f(x - y)\, dx d\nu = \int 1_A(x) \left(\int f(x - y)\, d\nu\right) dx$$

From this we make the following definition:

Definition. Let $f \in L^p$ and ν a probability measure. Then the convolution of f with ν is

$$f * \nu(x) = \int f(x - y)\, d\nu$$

Here we are taking the convolution of a density and a law rather than 2 laws. This actually gives us the density of $X+Y$, since in the above derivation we are saying that the probability that random variable is in A is $\int 1_A(x) \left(\int f(x - y)\, d\nu\right) dx$, and if we integrate with respect to x and for every A the probability is $\int 1_A(x) (h(x))\, dx$ then $h(x)$ is in fact the density. This follows from setting $g(x) = 1$ in our original density definition from earlier.

[1] The reason why we don't do much of the intuition in this sublevel is entirely the fault of the way that the Maths courses are designed, where one is meant to get the intuition somewhere and understand what is actually going on in a SEPARATE COURSE from when they get the Rigor and somehow put up with that! Often the intuition comes before the Rigor, which makes me feel like I am just doing magic, so I figure that by reordering the tripos courses on this website a little and doing it this way is the less of 2 evils. But nobody seems to like my idea of having the Rigor and intuition be in the same course so everyone can be happy, so for now you are gonna have to do this and wait until later for the motivation, and if you are like me you will be 10000000x happier than if we were to do it the other way around.

In particular, this implies that if we have two random variables that are independent and one of them has a density, then their sum has a density, given by $\int f(x-y)\,d\nu$.

We have to check that $f * \nu$ is actually a measurable function. For this it's Borel measurable by the first lemma in section 4.1.

Note that f does not need to integrate to 1 for this definition to work.

Proposition. If $f \in L^p$ and ν is a probability measure then $f * \nu$ is in L^p.

Proof. First consider the case where $p = \infty$. Then we are integrating a bounded a.e. thing with respect to a probability measure so we get something less than the bound for every x which is itself a bounded function. If p is finite then

$$\|f * \nu\|_p^p \leq \int \left(\int |f(x-y)|\,d\nu\right)^p dx$$

By Jensen's inequality this is

$$\begin{aligned}
&\leq \int\int |f(x-y)|^p\,d\nu dx \\
&= \int\int |f(x-y)|^p\,dx d\nu \\
&= \int |f(x-y)|^p\,dx < \infty
\end{aligned}$$

the last equality is because we are integrating with respect to a probability measure.

□

Corollary. By the above proof we have

$$\|f * \nu\|_p \leq \|f\|_p$$

Theorem. When we take the fourier transform of a convolution it terns into a product. Specifically,

$$\widehat{f * \nu}(u) = \hat{f}(u)\,\hat{\nu}(u)$$

Proof.

$$\widehat{f * \nu}(u) = \int \left(\int f(x-y)\,d\nu\right) e^{i(u.x)} dx$$

Since f is integrable and $|e^{i\text{anything}}| = 1$ we have integrability and so it is safe to swap the integrals around.

$$\begin{aligned}
&= \iint f(x-y)\,e^{i(u.x)} dx d\nu \\
&= \iint f(x-y)\,e^{i(u.(x-y))} e^{i(u.y)} d(x-y)\,d\nu \\
&= \iint f(x)\,e^{i(u.x)} dx e^{i(u.y)} d\nu \\
&= \iint f(x)\,e^{i(u.x)} dx e^{i(u.y)} d\nu = \int \hat{f}(u)\,e^{i(u.y)} d\nu \\
&= \hat{f}(u) \int e^{i(u.y)} d\nu = \hat{f}(u)\,\hat{\nu}(u)
\end{aligned}$$

□

In the case of probability measures, if X and Y are independent then we know the characteristic function of $X + Y$ is the product of the characteristic functions since $E\left[e^{i(X+Y)}\right] = E\left[e^{iX}\right] E\left[e^{iY}\right]$ by independence.

6.3 Fourier inversion

In this chapter we will show that the following is true if $f, \hat{f}$ is in L^1.

$$f(x) = \frac{1}{(2\pi)^d} \int \hat{f}(u) e^{-i(u.x)} du \text{ almost everywhere}$$

This gives us a way to recover f almost everywhere from its Fourier transform.

Definition. Define the density of a Gaussean with variance t as follows

$$g_t(x) = \left(\frac{1}{2\pi t}\right)^{\frac{d}{2}} e^{-\frac{|x|^2}{2t}}$$

This is equivalent to the density of d independent copies of $\sqrt{t}Z$ where Z is a standard normal.

We will show by direct computation that the Fourier inversion formula works for g_t.

First, let $d = 1$ and $Z \sim N(0, 1)$. Then we know by our proof of the central limit theorem in Level 6.3 that its characteristic function is $e^{-\frac{u^2}{2}}$.

Proposition. Let $g_t(x)$ be as above, then $\hat{g}_t(u) = e^{-\frac{|u|^2 t}{2}}$

Proof. Define Z to be a random vector with d independent copies of a standard normal.

$$\hat{g}_t(u) = E\left[e^{i\left(u.\sqrt{t}Z\right)}\right] = \prod E\left[e^{i\left(u_j\sqrt{t}Z_j\right)}\right] = \prod \phi_{N(0,1)}\left(\sqrt{t}u_j\right) = e^{-\frac{|u|^2 t}{2}}$$

□

In the rest of chapter 6.3 of this document all integrals written down are assumed to be over $\mathbb{R}^d$ unless stated otherwise.

Note that by direct computation we have

$$\hat{g}_t(u) = (2\pi)^{\frac{d}{2}} t^{-\frac{d}{2}} g_{\frac{1}{t}}(u)$$

This tells us that

$$\hat{\hat{g}}_t(u) = (2\pi)^d g_t(u)$$

Therefore we have

$$g_t(x) = g_t(-x) = (2\pi)^{-d} \hat{\hat{g}}_t(-x) = \left(\frac{1}{2\pi}\right)^d \int \hat{g}_t(u) e^{-i(u.x)} du$$

Therefore the fourier inversion formula works for the Gaussian density function.

Definition. Let $f \in L^1$. Then a Gaussian convolution of f is a function of the form $f * g_t$. We will consider this to represent convoluting f with the probability measure associated with the distribution with density g_t.

We will now do a bunch of inequalities on this which we will see why we need them after being done.

By earlier work we have

$$\|f * g_t\|_1 \leq \|f\|_1$$

We also have the bound

$$|f * g_t(x)| = \left|\int f(x-y)\, dg_t\right| = \left|\int f(x-y) e^{-\frac{|y|^2}{2t}} \left(\frac{1}{2\pi t}\right)^{\frac{d}{2}} dy\right|$$

$$\leq \left(\frac{1}{2\pi t}\right)^{\frac{d}{2}} \int \left|f(x-y) e^{-\frac{|y|^2}{2t}}\right| dy \leq \left(\frac{1}{2\pi t}\right)^{\frac{d}{2}} \int |f(x-y)|\, dy \leq \left(\frac{1}{2\pi t}\right)^{\frac{d}{2}} \|f\|_1$$

This tells us that

$$\|f * g_t\|_\infty \leq \left(\frac{1}{2\pi t}\right)^{\frac{d}{2}} \|f\|_1$$

This means that the convolution of $f * g_t$ is bounded for each t but the bound gets worse as t gets smaller, which we expect because we did not assume that $f * g_0 = f$ is bounded (Note that g_0 is not well defined but that is ok because this is just an intuition remark and not a claim).

We also have

$$\left\|\widehat{f * g_t}\right\|_1 = \left\|\hat{f}\hat{g}_t\right\|_1 = \int \left|\hat{f}\left(u\right)\hat{g}_t\left(u\right)\right| du \leq \left\|\hat{f}\right\|_\infty \int \left|\hat{g}_t\left(u\right)\right| du = \left\|\hat{f}\right\|_\infty \|\hat{g}_t\|_1$$

Note that by an argument using the triangle inequality for integrals, the fourier transform of an L^1 function is bounded by the L^1 norm of the original function. This gives, from the line above,

$$\left\|\widehat{f * g_t}\right\|_1 \leq \|f\|_1 \|\hat{g}_t\|_1 = (2\pi)^{\frac{d}{2}} t^{-\frac{d}{2}} \|f\|_1$$

We have, also

$$\left\|\widehat{f * g_t}\right\|_\infty = \left\|\hat{f}\hat{g}_t\right\|_\infty \leq \left\|\hat{f}\right\|_\infty \|\hat{g}_t\|_\infty$$

The last inequality is justified because not only are suprema of product bounded by products of suprema, but the problem of the infinity norm only dealing with the "almost everywhere" case, which has actually been fine in previous computations, is fine here as well by continuity of Fourier transforms. Since $\hat{g}_t$ is something we already computed and is bounded by 1, we get the inequality

$$\left\|\widehat{f * g_t}\right\|_\infty \leq \left\|\hat{f}\right\|_\infty$$

We are going to try to prove the Fourier inversion formula for $f * g_t$. These bounds check that $f * g_t$ and its fourier transform is actually in L^1. We don't technically need the infinity norm bounds but they are useful "consistency checks" as fourier transforms of functions are known to be bounded by earlier work by the L^1 norm of the original function.

Now let's actually do the proof.

Proposition. The Fourier inversion formula holds for $f * g_t$

Proof.

$$f * g_t\left(x\right) = \int f\left(x-y\right) g_t\left(y\right) dy$$

$$= \int f\left(x-y\right)\left(\frac{1}{(2\pi)^d}\int \hat{g}_t\left(u\right) e^{-i(u.y)} du\right) dy$$

$$= \frac{1}{(2\pi)^d} \iint \left(f\left(x-y\right)\hat{g}_t\left(u\right) e^{-i(u.y)}\right) dudy$$

We're in L^1 so we can swap integrals around.

$$= \frac{1}{(2\pi)^d} \int \left(\int f\left(x-y\right) e^{i(u.(x-y))} dy\right) \hat{g}_t\left(u\right) e^{-i(u.x)} du$$

$$= \frac{1}{(2\pi)^d} \int \hat{f}\left(u\right)\hat{g}_t\left(u\right) e^{-i(u.x)} du$$

$$= \frac{1}{(2\pi)^d} \int \widehat{f * g_t}\left(u\right) e^{-i(u.x)} du$$

□

Note that a convolution is like adding random variables. We therefore expect that $f * g_t$ should be a good approximation for f as $t \to 0$. We will prove the lemma "Let $f \in L^p$ and suppose that p is finite. Then $\|f * g_t - f\|_p \to 0$ as $t \to 0$" shortly but first we need another lemma.

Lemma. Let $f \in L^p$ and suppose that p is finite and let $\varepsilon > 0$. Then there exists a continuous function h which is 0 except on a bounded set (i.e. has compact support) such that $\|f - h\|_p \leq \varepsilon$.

Proof. We can assume f is non-negative by proving the theorem for the positive and negative parts of f and adding the h functions together.

Since f is non-negative measurable, by general analysis there exists a sequence of non-negative simple functions s_n that increase pointwise to f and the dominated convergence theorem gurantees that $s_n \to f$ in L^p. In particular we can pick a simple function s such that $\|f - s\|_p < \frac{\varepsilon}{3}$. Lets say that this simple function is $\sum_{i=1}^m a_i 1_{A_i}$.

The measurable sets A_i have finite measure but that does not mean that they are actually bounded. Intersect it with the ball $B_R(0)$ and note that by continuity of measures we have $\mu(A_i \cap B_R(0)) \to \mu(A_i)$ as $R \to \infty$. Therefore we can find an R such that $\|s - s_R\|_p < \frac{\varepsilon}{3}$. Call this $A_i \cap B_R(0)$ set E_i.

By properties of the Lebesgue measure which we proved at the beginning of the course, for every bounded measurable set E and every $\delta > 0$ there exists a compact set K and an open set U such that $K \subseteq E \subseteq U$ and $\mu(U \setminus K) < \delta$. We can pick U to be bounded because E is bounded.

We will construct a continuous function $g : \mathbb{R}^d \to \mathbb{R}$ as follows using the "distance to boundary of a closed set or 0 if in the closed set" trick:

$$g(x) := \frac{d(x, U^C)}{d(x, U^C) + d(x, K)}$$

Note that this is 1 in K and K and U^c are disjoint closed sets so the denominator is never 0. Furthermore, it is 1 everywhere in K and 0 everywhere in U^C and continuous.

Clearly, the p-norm of $1_E - g$ will be $< \delta$. We can now assemble our final function. For each E_i find a coresponding function g_i such that

$$\|1_{E_i} - g_i\|_p < \frac{\varepsilon}{3ma_i}$$

Define $h = \sum_{i=1}^m a_i g_i$

Using Minkowski's inequality it follows that h is the function we want.

□

Lemma. Let $f \in L^p$ and suppose that p is finite. Then $\|f * g_t - f\|_p \to 0$ as $t \to 0$

Note that if $p = \infty$ this is false in general because if f is not continuous this will contradict the uniform limit of continuous functions being continuous, and $f * g_t$ is continuous because by the above proof it is expressible as a fourier transform if we replace u with $-u$.

Proof. Fix $\varepsilon > 0$, then by the previous lemma we can find a continuous function h with compact support such that $\|f - h\|_p \leq \frac{\varepsilon}{3}$. Therefore we have

$$\|f * g_t - h * g_t\|_p = \|(f - h) * g_t\|_p \leq \|f - h\|_p \leq \frac{\varepsilon}{3}$$

Therefore we only need to prove the lemma for continuous functions with compact support. Specifically, we have

$$\|f * g_t - f\|_p \leq \|f * g_t - h * g_t\|_p + \|f - h\|_p + \|h * g_t - h\|_p \leq \frac{2\varepsilon}{3} + \|h * g_t - h\|_p$$

So we just need to make the last term $< \frac{\varepsilon}{3}$. Define

$$e(y) = \int |h(x - y) - h(x)|^p \, dx$$

Then e is bounded (it is here where we use the fact that p is finite) because

$$e(y) \leq 2^p \int |h(x - y)|^p + |h(x)|^p \, dx = 2^{p+1} \|h\|_p^p$$

Since h is continuous and bounded and has compact support the dominated convergence theorem implies that $e(y) \to 0$ as $y \to 0$.

We have

$$\begin{aligned}\|h * g_t - h\|_p^p &= \int \left| \int h(x-y) g_t(y)\, dy - h(x) \right|^p dx \\ &= \int \left| \int (h(x-y) - h(x)) g_t(y)\, dy \right|^p dx \\ &\leq \int \int |h(x-y) - h(x)|^p g_t(y)\, dy dx\end{aligned}$$

The last inequality is by Jensen's inequality since $g_t(y)\, dy$ is a probability measure. By non-negative-ness of everything we are safe to write

$$\begin{aligned}&= \int e(y) g_t(y)\, dy \\ &= \int e\left(\sqrt{t} y\right) g_1(y)\, dy\end{aligned}$$

Note that this is just integration by substitution on the last line with the change from g_t to $g_1$absorbing the changed factor.

By the dominated convergence theorem this goes to 0 as t goes to 0 so we are done.

□

Now we will prove the Fourier inversion formula.

Proof. Recall that $\widehat{f * g_t}(u) = \hat{f}(u) e^{-\frac{|u^2|t}{2}}$. Define $f_t := f * g_t$, then this satisfies the fourier inversion formula. Specifically, we have

$$f_t(x) = (2\pi)^{-d} \int \hat{f}(u) e^{-\frac{|u^2|t}{2}} e^{-i(u.x)} du$$

We know that the absolute value of the integrand is bounded above by $\left|\hat{f}\right|$ which by assumption is in L^1 so by the dominated convergence theorem with dominating function $\left|\hat{f}\right|$ and letting $t \to 0$, we have

$$\lim_{t \to 0} f_t(x) = (2\pi)^{-d} \int \hat{f}(u) e^{-i(u.x)} du$$

Since $f_t \to f$ in measure, by a theorem we proved previously there is a sequence (t_n) tending to 0 such that $f_{t_n} \to f$ almost everywhere. (Note: It is here that the "almost everywhere" part of the theorem comes in, at the very end) Combining these, we have that almost everywhere,

$$f(x) = (2\pi)^{-d} \int \hat{f}(u) e^{-i(u.x)} du$$

so done

□

6.4 Fourier transform in L^2

Theorem. (Plancherel identity) For any function $f \in L^1 \cap L^2$ we have the following

$$\left\|\hat{f}\right\|_2 = (2\pi)^{\frac{d}{2}} \|f\|_2$$

Note that $L^1 \cap L^2$ is necessary as $\frac{1}{1+|x|}$ is in L^2 but not L^1 and the function $\frac{1}{\sqrt{x}}$ for x in $(0,1)$ and 0 elsewhere is in L^1 but not in L^2.

Proof. First assume $\hat{f}$ is in L^1 so that we can use the Fourier inversion formula.

$$\|f\|_2^2 = \int f(x)\overline{f(x)}dx$$

$$\|f\|_2^2 = \frac{1}{(2\pi)^d}\int\int \hat{f}(u)e^{-i(u.x)}du\overline{f(x)}dx$$

$$\|f\|_2^2 = \frac{1}{(2\pi)^d}\int \hat{f}(u)\left(\int \overline{f(x)}e^{-i(u.x)}dx\right)du$$

$$\|f\|_2^2 = \frac{1}{(2\pi)^d}\int \hat{f}(u)\left(\overline{\int f(x)e^{i(u.x)}dx}\right)du$$

$$\|f\|_2^2 = \frac{1}{(2\pi)^d}\int \hat{f}(u)\overline{\hat{f}(u)}du$$

Now let $f_t = f * g_t$. Then by earlier work we know that $\|f_t\|_2 \to \|f\|_2$ as $t \to 0$. Now note that

$$\hat{f}_t(u) = \hat{f}(u)\hat{g}_t(u) = \hat{f}(u)e^{-\frac{|u|^2t}{2}}$$

As $t \to 0$, $e^{-\frac{|u|^2t}{2}} \to 1$ so we know by the monotone convergence theorem that

$$\left\|\hat{f}_t\right\|_2^2 = \int \left|\hat{f}(u)\right|^2 e^{-|u|^2t}du \to \int \left|\hat{f}(u)\right|^2 du = \left\|\hat{f}\right\|_2^2$$

Since $\|f * g_t\| \leq \|f\|$ by earlier work on convolutions, $f_t \in L^1$. We also have

$$\left\|\hat{f}_t\right\|_1 = \int \left|\hat{f}(u)\right| e^{-|u|^2t}du \leq \|f\|_1 \int e^{-|u|^2t}du < \infty$$

Thus $\hat{f}_t \in L^1$. Therefore the identity holds for f_t, which means that

$$\left\|\hat{f}_t\right\|_2 = (2\pi)^{\frac{d}{2}}\|f_t\|_2$$

The result follows as we proved we can take the limit as $t \to 0$.

□

From now on, we denote $\mathcal{L}^2$ to be the set of equivalence classes of L^2.

Definition. We say a function between Hilbert spaces is an automorphism if it preserves the inner product, is linear, and is a bijection.

Theorem. There exists a unique Hilbert space automorphism $F : \mathcal{L}^2 \to \mathcal{L}^2$ such that for all $f \in L^1 \cap L^2$

$$F([f]) = \left[(2\pi)^{-d/2}\hat{f}\right]$$

Proof. We define a function $F_0 : \mathcal{L}^1 \cap \mathcal{L}^2 \to \mathcal{L}^2$ by

$$F_0([f]) = \left[(2\pi)^{-\frac{d}{2}}\hat{f}\right]$$

This preserves the L^2 norm and is linear by the Plancherel identity.

We proved earlier that continuous functions with compact support are dense in $\mathcal{L}^2$ and therefore $\mathcal{L}^1 \cap \mathcal{L}^2$ is dense in $\mathcal{L}^2$. To extend F_0 to general functions in $\mathcal{L}^2$ we simply take the limit of a sequence of functions in $\mathcal{L}^1 \cap \mathcal{L}^2$. Note that if f_n is a Cauchy sequence in $\mathcal{L}^1 \cap \mathcal{L}^2$ then so is $F_0(f_n)$, since F_0 preserves the L^2 norm. Also, the limit function F is unique since limits are unique in this space as it is Hausdorff. This means that if there exists such an extension it is unique. It remains to show that it is actually an automorphism.

To show that it is linear, we use linearity of F_0. Specifically,

$$F(ax+by) = \lim_{n\to\infty} F_0(ax_n + by_n)$$

Note that $ax_n + by_n$ is in the domain of F_0 since F_0 is a vector space. Then we have

$$= \lim_{n\to\infty} \left(aF_0(x_n) + bF_0(y_n)\right) = aF(x) + bF(y)$$

To get that F is an isometry, we want to show that $\|F(x)\|_2 = \|x\|_2$ for all $x \in L^2$. We can choose a sequence x_n in $\mathcal{L}^1 \cap \mathcal{L}^2$ such that $x_n \to x$. The norm function is uniformly continuous because

$$|\|u\| - \|v\|| \leq \|u - v\|$$

Therefore we can pass limits inside the norm. Then we have by F_0's isometry that

$$\|F(x)\|_2 = \left\|\lim_{n\to\infty} F_0(x_n)\right\|_2 = \lim_{n\to\infty} \|F_0(x_n)\|_2 = \lim_{n\to\infty} \|x_n\|_2 = \|x\|_2$$

Now we just need to show that F is injective and surjective. It is injective since it preserves distance. To prove that it is surjective, note that the map

$$V = \left\{[f] \in \mathcal{L}^2 : f, \hat{f} \in L^1\right\}$$

is sent to itself, and $F^4[f] = [f]$ (since $F^2[f](x) = [f](-x)$), so V is contained in the image of F and dense in $\mathcal{L}^2$ since it contains all Gaussian convolutions by earlier work and this implies density by earlier work. Therefore F is surjective since it contains all limit points of a dense set.

□

6.5 Characteristic functions

We proved in Level 6.3 that they determine the distribution and that convergence of characteristic functions pointwise is equivalent to convergence in distribution. However, there is one more theorem which we will prove here.

Theorem. Let ϕ_X be the characteristic function of a random variable X and suppose that it is integrable. Then X has a bounded, continuous probability density function and its density is equal to

$$f_X = (2\pi)^{-d} \int \phi_X(u)\, e^{-i(u.x)} du$$

Proof. Let $Z \sim N(0,1)$ be independent of X. Then consider $X + \sqrt{t}Z$. Since Fourier inversion holds for Gaussian convolutions we have

$$f_{X+\sqrt{t}Z}(x) = (2\pi)^{-d} \int \phi_X(u)\, e^{\frac{|u|^2 t}{2}} e^{-i(u.x)} du$$

This is because $X + \sqrt{t}Z$ has a density by our work in the part on Convolutions.

Letting t go to 0 and using dominated convergence theorem with dominating function $|\phi_X|$ the result follows.

We now want to check that $\lim_{t\to 0} f_{X+\sqrt{t}Z}(x)$ is bounded, continuous and actually a density.

It is bounded by $(2\pi)^{-d} \int |\phi_X(u)|\, du$. It is here we use the fact that the characteristic function is integrable.

It is continuous for the same reason that Fourier transforms are continuous.

As $t \to 0$, for almost every ω in the sample space, $X(\omega) + \sqrt{t}Z(\omega) \to X(\omega)$ so we have almost sure convergence, thus convergence in probability. For any bounded continuous function g with compact support we have $g\left(X + \sqrt{t}Z\right) \to g(X)$ almost surely, and thus also in probability. By the bounded convergence theorem, the sequence $g\left(X + \sqrt{t}Z\right)$ converges to $g(X)$ in L^1, and thus $E\left[g\left(X + \sqrt{t}Z\right)\right] \to E[g(X)]$. By the definition of the density of f_t we have $E\left[g\left(X + \sqrt{t}Z\right)\right] =$

$\int g(x) f_t(x)\, dx$. Since g is compactly supported and f_t is uniformly bounded by a constant independent of t, we can apply the dominated convergence theorem to get that

$$\int g(x) f_t(x)\, dx \to \int g(x) f(x)\, dx$$

Therefore the limits we need to coincide hold, and f is a density since this holds for all such functions g. By the monotone convergence theorem and a sequence of continuous functions g increasing to indicator functions of intervals, it holds if g is such a function. By the monotone class theorem and the same "taking a max with n" trick, it works for g if g is non-negative measurable. Therefore f is indeed a density.

□

As we saw in Level 6.3, you can prove that characteristic functions determine the distribution and that convergence of characteristic functions implies convergence in distribution. With these tools in hand, you can prove the central limit theorem.

6.6 The multivariate normal

In the IA probability course (level 8.4) we proved most of the following properties of the multivariate normal, or at least, I did in my notes.

Definition. A multivariate normal is a random vector such that all its projections are 1D normals.

Theorem. Let X be a multivariate normal on $\mathbb{R}^n$ and let A be an $m \times n$ matrix and $b \in \mathbb{R}^m$. Then we have

i) $AX + b$ is normal on $\mathbb{R}^m$

ii) $X \in L^2$

iii) The distribution of X is determined by its mean μ and covariance matrix V

iv) Its characteristic function is given by $e^{i(u-\mu)-\frac{u^T V u}{2}}$

v) If V is invertible then the density of X is

$$(2\pi)^{-\frac{n}{2}} (\det(V))^{-\frac{1}{2}} \exp\left(-\frac{1}{2}\left((x-\mu)^T \left(V^{-1}\right)(x-\mu)\right)\right)$$

vi) If $X = (X_1, X_2)$ where $X_i \in \mathbb{R}^{n_i}$ then $\operatorname{cov}(X_1, X_2) = 0$ if and only if they are independent.

Proof. i, iii and iv were proven in the IA probability course. For (ii) note that we are done since each component is in L^2. For (iv) note that we just need to evaluate the moment generating function which we computed in IA probability at iu. For (vi) we note that independence implies 0 covariance, and by a similar argument to in IA probability using properties of moment generating functions we get the converse.

□

7 Ergodic theory

7.1 Definitions and basic properties

In this section we will be working in a measure space $(E, \mathcal{E}, \mu)$ where we have a measurable map $\Theta : E \to E$ that has the property that for all $A \in \mathcal{E}$ we have $\mu(A) = \mu\left(\Theta^{-1}(A)\right)$.

Example. On $(0, 1)$, the map $(x + a) \bmod 1$ would satisfy the above constraint.

For a measurable function f we will define $S_n(f) = f + f \circ \Theta + f \circ \Theta \circ \Theta + ... + f \circ \Theta^{n-1}$. We then want to understand the long run behaviour of $\frac{S_n(f)}{n}$ as $n \to \infty$.

Definition. $A \in \mathcal{E}$ is said to be invariant for Θ if $A = \Theta^{-1}(A)$

Definition. A measurable function f is said to be invariant if $f = f \circ \Theta$

Definition. We write $\mathcal{E}_\Theta = \{A \in \mathcal{E} : A \text{ is invariant}\}$. Then by checking against the definition we see that this is a σ-algebra.

Proposition. A measurable function $f : E \to \mathbb{R}$ is invariant if and only if it is $\mathcal{E}_\Theta$-measurable

Proof. If the function is indeed invariant then pre-images of all measurable sets under f will be the same as the pre-images of all measurable sets under $f \circ \Theta$, so $f^{-1}(A) = \Theta^{-1}\left(f^{-1}(A)\right)$. But this is an equivalent condition to saying that the pre-image of all measurable sets are invariant sets.

□

Definition. We say Θ is ergodic if $A \in \mathcal{E}_\Theta$ implies that $\mu(A) = 0$ or $\mu\left(A^C\right) = 0$.

Example. For the translation map on $[0, 1)$ it is easy to check that it is not ergodic if a is rational, as if $a = \frac{p}{q}$ any $\frac{1}{q}$-periodic set will do. If a is irrational this is actually an interesting, difficult question.

Proposition. If f is integrable and Θ is measure-preserving then $f \circ \Theta$ is integrable and $\int_E f \circ \Theta d\mu = \int_E f d\mu$

Proof. It is measurable because pre-images of measurable sets remain measurable by definition and the integrals are equal because if we have a simple function underestimating f^+ then it corresponds to a simple function underestimating $f^+ \circ \Theta$ with the same measure and this is basically the idea.

□

Proposition. If Θ is ergodic and f is invariant then there exists a constant c such that $f = c$ almost everywhere

Proof. Define the set $A_c = \{x : f(x) < c\}$ for each c. Then this is an invariant set which means by ergodic-ness it must either have measure 0 or its complement much have measure 0. Then we would expect c to be the supremum of the set where the measure of A_c is 0 and if there is no such c then f would have to be infinite almost everywhere which is a contradiction.

□

7.2 The shift map

Let m be a probability distribution on $\mathbb{R}$. Then there exists an iid sequence $Y_1, Y_2, ...$ with law m. We will let $E = \mathbb{R}^n$ be the set of all real sequences (x_n). We define the σ-algebra $\mathcal{E}$ to be the σ-algebra generated by all the projections $X_n(x) = x_n$, which means the smallest σ-algebra such that all of these functions are measurable. In other words, it is the σ-algebra generated by the set

$$\left\{\prod_{n\in\mathbb{N}} A_n, A_n = \mathbb{R} \text{ for all n except one k for which } A_k \in \mathcal{B}\right\}$$

Which makes it generated by the π-system

$$\mathcal{A} = \left\{\prod_{n\in\mathbb{N}} A_n, A_n \in \mathcal{B} \text{ for all n and } A_n \in \mathbb{R} \text{ eventually}\right\}$$

To define the measure μ we let $Y = (Y_1, Y_2, ...) : \Omega \to E$ where Y_i are iid and Ω is their sample space. Then this is a measurable map because each of the Y_i's is a random variable and we will let $\mu = P \circ Y^{-1}$. By independence of the Y_i's, we have that for any $A = A_1 \times A_2 \times ... \times A_n \times \mathbb{R} \times ... \times \mathbb{R} \times ...$ we have that $\mu(A) = \prod_{n\in\mathbb{N}} m(A_n)$. This is really a finite product since the terms are eventually all 1.

This is then known as the canonical space associated with the sequence of iid random variables with law m. We will define the shift map $\Theta : E \to E$ by $\Theta(x_1, x_2, x_3, ...) = (x_2, x_3, x_4, ...)$. Hopefully you start to see now why this relates to the strong law of large numbers. This is useful because if $f(x) = x_1$ then $S_n(f) = x_1 + ... + x_n$. Ergodic theory will then tell us about the long-run behaviour of the average.

Theorem. The shift map Θ as defined above is an Ergodic, measure preserving transformation

Proof. Clearly it is measurable. It is also measure preserving because the pre-image of $(A_1, A_2, ...)$ is $(\mathbb{R}, A_1, A_2, ...)$ which has the same measure.

Recall the definition of a tail σ-algebra T from earlier. Suppose that $A \in \prod_{n\in\mathbb{N}} A_n \in \mathcal{A}$, then $\Theta^{-n}(A) = \{X_{n+k} \in A_k \text{ for all } K\} \in T_n$.

Since T_n is a σ-algebra and $\Theta^{-n}(A) \in T_n$ for all $A \in \mathcal{A}$ and $\sigma(\mathcal{A}) = \mathcal{E}$, we know that $\Theta^{-n}(A) \in T_n$ for all $A \in \mathcal{E}$.

Therefore if $A \in \mathcal{E}_\Theta$ i.e. $A = \Theta^{-1}(A)$ then $A \in T_N$ for all N so $A \in T$.

This means that by the Kolmogorov 0-1 law, either $\mu(A) = 1$ or $\mu(A) = 0$. This means that Θ is ergodic.

$\square$

7.3 Ergodic theorems

Lemma. (Maximal ergodic lemma) Let f be integrable and let $S^* = \sup_{n\geq 0} S_n(f)$. We define $S_0(f) = 0$ by convention which implies $S^* \geq 0$. Then the claim is that

$$\int_{\{S^*>0\}} f d\mu \geq 0$$

Proof. Define $S_n^* := \max_{0\leq m\leq n} S_m$ and $A_n := \{S_n^* > 0\}$. If $1 \leq m \leq n$ then we know that

$$S_m = f + S_{m-1} \circ \Theta \leq f + S_n^* \circ \Theta$$

Since $S_0 = 0$ we have that on A_n, $S_n^* = \max_{1\leq m\leq n} S_m$. We therefore have

$$S_n^* \leq f + S_n^* \circ \Theta$$

On A_n^C we have

$$S_n^* = 0 \leq S_n^* \circ \Theta$$

Therefore we know that

$$\begin{aligned} \int_E S_n^* d\mu &= \int_{A_n} S_n^* d\mu + \int_{A_n^C} S_n^* d\mu \\ &\leq \int_{A_n} f d\mu + \int_{A_n} S_n^* \circ \Theta d\mu + \int_{A_n^C} S_n^* \circ \Theta d\mu \\ &= \int_{A_n} f d\mu + \int_E S_n^* \circ \Theta d\mu = \int_{A_n} f d\mu + \int_E S_n^* d\mu \end{aligned}$$

It therefore follows that $\int_{A_n} f d\mu \geq 0$. By dominated convergence with dominating function f, the result follows.

$\square$

Theorem. (Birkhoff's ergodic theorem) Let $(E, \mathcal{E}, \mu)$ be σ-finite and let $f : E \to \mathbb{R}$ be integrable. Then there exists an invariant function such that

$$\mu\left(\left|\bar{f}\right|\right) \leq \mu\left(|f|\right)$$

and

$$\frac{S_n(f)}{n} \to \bar{f} \text{ a.e.}$$

And if Θ is ergodic then $\bar{f}$ is a constant.

Proof. Note that $\limsup_n \frac{S_n}{n}$ and $\liminf_n \frac{S_n}{n}$ are invariant functions. This is because $S_n \circ \Theta = S_{n+1} - f$ so we have

$$\limsup_{n\to\infty} \frac{S_n \circ \Theta}{n} = \limsup_{n\to\infty} \frac{S_{n+1}}{n} - \frac{f}{n} = \limsup_{n\to\infty} \frac{S_n}{n+1} = \limsup_{n\to\infty} \frac{S_n}{n}$$

and similarly for liminf.

We will now try to show that the set of points on which limsup and liminf do not agree has measure 0. Set $a < b$ and define

$$D(a,b) = \left\{x \in E : \liminf_{n\to\infty} \frac{S_n}{n} < a < b < \limsup_{n\to\infty} \frac{S_n}{n}\right\}$$

If the limsup and liminf are not equal we can find rational numbers a, b such that $x \in D(a,b)$. By countable subadditivity we therefore will show that $\mu(D(a,b)) = 0$ for all a, b. Now fix a, b and just call the set D. Since limsup and liminf are invariant, we have that D is also invariant. By restricting to D we can assume that $D = E$. Suppose that $B \in \mathcal{E}$ and $\mu(B) < \infty$ and let $g := f - b1_B$. Then g is integrable because f is integrable and $\mu(B) < \infty$ and we have

$$S_n(g) = S_n(f - b1_B) \geq S_n(f) - nb$$

By definition we know that $\limsup_{n\to\infty} \frac{S_n(f)}{n} > b$ so we can find an n with $S_n(g) > 0$. Then we know that $S^*(g)(x) > 0$ for all $x \in D$ where S^* is as defined earlier. So by the maximal ergodic lemma we have that

$$0 \leq \int_D g d\mu = \int_D (f - b1_B) d\mu = \int_D f d\mu - b\mu(B)$$

This therefore implies that

$$b\mu(B) \leq \int_D f d\mu$$

This holds for all $b \in \mathcal{E}$ with finite measure. Since our space is σ-finite we can find a sequence $B_n \nearrow D$ such that $\mu(B_n) < \infty$ for all n. Taking the limit above gives

$$b\mu(D) \leq \int_D f d\mu$$

We can run the same argument with $-a$ in place of b and $-f$ in place of f to get

$$a\mu(D) \geq \int_D f d\mu$$

Note that since $b > a$, at least one of $b > 0$ and $a < 0$ has to be true. If $b > 0$ then integrability of f implies that $\mu(D)$ is finite. In this case we have

$$b\mu(D) \leq \int_D f d\mu \leq a\mu(D)$$

But $a < b$ so this can only happen if $\mu(D) = 0$ which is what we want. If $a < 0$ then again $\mu(D)$ is finite so we can run the same argument.

This means that we have proven that the set of points on which $\liminf_{n\to\infty} \frac{S_n}{n} \neq \limsup_{n\to\infty} \frac{S_n}{n}$ has measure 0.

We now define

$$\bar{f}(x) = \begin{cases} 0 & x \in D \\ \lim\left(\frac{S_n(f)}{n}\right) & x \in E \setminus D \end{cases}$$

Then $\frac{S_n(f)}{n} \to \bar{f}$ almost everywhere. It remains to prove that $\mu\left(\left|\bar{f}\right|\right) \leq \mu(|f|)$ and that if Θ is ergodic then $\bar{f}$ is constant.

Note that since we have a measure-preserving function, $\mu(|f \circ \Theta^n|) = \mu(|f|)$ and therefore $\mu(|S_n|) \leq n\mu(|f|) < \infty$. By Fatou's lemma, we have

$$\mu\left(\left|\bar{f}\right|\right) \leq \mu\left(\liminf_n \left|\frac{S_n(f)}{n}\right|\right) \leq \liminf_n \mu\left(\left|\frac{S_n(f)}{n}\right|\right) = \mu(|f|)$$

If Θ is ergodic then by a previous theorem an invariant function can be made to be constant almost everywhere so if we set $\bar{f}$ to be that constant then it will work.

□

Theorem. (Von Neumann's ergodic theorem) Let $(E, \mathcal{E}, \mu)$ be a finite measure space. Let $p \in [1, \infty)$ and let $f \in L^p$. Then there exists a function $\bar{f} \in L^p$ such that $\frac{S_n(f)}{n} \to \bar{f}$ in L^p.

Proof. With the same proof as earlier, we have

$$\|f \circ \Theta\|_p^p = \int |f \circ \Theta|^p \, d\mu = \int |f|^p \, d\mu = \|f\|_p^p$$

Therefore we have

$$\left\|\frac{S_n}{n}\right\|_p = \frac{1}{n}\left\|f + f \circ \Theta + f \circ \Theta \circ \Theta + \ldots + f \circ \Theta^{n-1}\right\| \leq \|f\|_p$$

by Minkowski's inequality.

Now fix $\varepsilon > 0$ and take $M \in (0, \infty)$ large enough such that if $g = (f \vee (-M)) \wedge M$ then $\|f - g\|_p < \frac{\varepsilon}{3}$ and thus $\left\|\frac{S_n(f-g)}{n}\right\|_p < \frac{\varepsilon}{3}$. By Birkhoff's theorem we know that $\frac{S_n(g)}{n} \to \bar{g}$ almost everywhere. Also, we know that $\left|\frac{S_n(g)}{n}\right| \leq M$ for all n so by the bounded convergence theorem we know that

$$\left\|\frac{S_n(g)}{n} - \bar{g}\right\|_p \to 0$$

as $n \to \infty$ and therefore we can find N such that $n > N$ implies $\left\|\frac{S_n(g)}{n} - \bar{g}\right\|_p < \frac{\varepsilon}{3}$. Then we have

$$\begin{aligned} \left\|\bar{f} - \bar{g}\right\|_p &= \int \liminf_n \left|\frac{S_n(f-g)}{n}\right|^p d\mu \\ &\leq \liminf_n \int \left|\frac{S_n(f-g)}{n}\right|^p d\mu \leq \|f - g\|_p^p \end{aligned}$$

Therefore if n is sufficiently large we have have

$$\left\|\frac{S_n(f)}{n} - \bar{f}\right\|_p \leq \left\|\frac{S_n(f-g)}{n}\right\|_p + \left\|\frac{S_n(g)}{n} - \bar{f}\right\|_p + \left\|\bar{g} - \bar{f}\right\|_p < \varepsilon$$

□

7.4 The strong law of large numbers

Recall that in IA probability we proved the strong law of large numbers under the assumption that $E\left[|X_1|^4\right]$ is finite. Here we will prove it assuming only that $E\left[|X_1|\right]$ is finite. The weak law of large numbers says $\frac{S_n}{n} \to \mu$ in probability if $E\left[\left|X_1^2\right|\right]$ is finite so that is different. The strong law is indeed stronger as a.s. convergence implies convergence in probability and our condition is weaker.

Theorem. (The strong law of large numbers, part 1) Let $X_1, X_2, ...$ be independent and have the same mean and suppose there exists M with $E\left[|X_n|^4\right] < M$ for all n, then $\frac{\sum_{i=1}^n X_i}{n}$ converges almost surely to $E\left[X_1\right]$ which we will call μ.

Proof. Note that for all $i \neq j$ we have, by Jensen's inequality, $E\left[X_i^2 X_j^2\right] = E\left[X_i^2\right] E\left[X_j^2\right] \leq \sqrt{E\left[X_i^4\right] E\left[X_j^4\right]} \leq M$

Then the rest of the proof proceeds exactly as in IA probability.

□

Theorem. (The strong law of large numbers, part 2) Let $X_1, X_2, ...$ be iid such that $E\left[|X_1|\right]$ is finite, then $\frac{\sum_{i=1}^n X_i}{n}$ converges almost surely to $E\left[X_1\right]$ which we will call μ. It also converges in L^1 to μ.

Proof. Let m be the law of X_1 and let $X = (X_1, X_2, X_3, ...)$ viewed as a function $X : \Omega \to \mathbb{R}^{\mathbb{N}} = E$. Then we let $(E, \mathcal{E}, \mu)$ be the canonical space associated with the distribution m so that $\mu = P \circ Y^{-1}$ and we let $f : E \to \mathbb{R}$ be given by $f\left(x_1, x_2, ...\right) = x_1$.

Then X_1 has law m and is integrable, and the shift map $\Theta\left(x_1, x_2, ...\right) = \left(x_2, x_3, ...\right)$ is measure-preserving and ergodic. Thus, by Birkhoff's theorem, $\frac{\sum_{i=1}^n X_i}{n} = \frac{S_n(f)}{n}$ converges almost surely to a constant $\bar{f}$ and we have convergence in L^1 by the von Neumann ergodic theorem.

To find the constant $\bar{f}$ we note that

$$\bar{f} = \mu\left(\bar{f}\right) = \lim_{n\to\infty} \mu\left(\frac{S_n\left(f\right)}{n}\right) = \mu$$

□